

Objectives

- Build a reproducible and auditable market churn analytics pipeline.
- Build a churn prediction and market analytics model with >85% accuracy and ROC-AUC >0.85.
- Use advanced machine learning methods such as Random Forest, XGBoost, and Ensemble techniques for reliable performance.
- Visualize churn risk and interventions through an operational dashboard.
- Quantify ROI of targeted retention actions to support strategic business decisions.

Dataset Overview

- **Primary Source:** Telecom Regulatory Authority of India (TRAI) — official telecom subscription data.
- **Dataset Size:** ~70,728 records (~25+ MB).
- **Time Span:** 2009–2025 (15+ years of historical data).
- **Coverage:**
 - 42 unique telecom operators across **38 telecom circles/states** in India.
 - Includes both **wireless and wireline** connection types.
- **Granularity:** Monthly data capturing subscriber counts by operator, circle, and technology (2G/3G/4G/5G).
- Reflects **telecom industry evolution** — from early 2G networks to 4G and 5G adoption.
- Facilitates **regional comparisons**, highlighting strong and weak circles for each operator.
- Serves as a **foundation for churn, market share, and ROI analysis** through feature engineering and predictive modeling.

	year	month	circle	type_of_connection	service_provider	value	unit	notes
0	2025	April	Andhra Pradesh	wireless	Bharti Airtel (Including Tata Tele.)	33965795	value in absolute number	NaN
1	2025	April	Assam	wireless	Bharti Airtel (Including Tata Tele.)	12314102	value in absolute number	NaN
2	2025	April	Bihar	wireless	Bharti Airtel (Including Tata Tele.)	40967773	value in absolute number	NaN
3	2025	April	Delhi	wireless	Bharti Airtel (Including Tata Tele.)	18877637	value in absolute number	NaN
4	2025	April	Gujarat	wireless	Bharti Airtel (Including Tata Tele.)	12401101	value in absolute number	NaN

Telecom Subscriptions,
provided dataset.

Exploratory Data Analysis

Understanding the structure, quality, and trends in the dataset before applying any advanced modeling

Data Cleaning :

- **Handled Missing Values:** Checked for nulls and removed or imputed missing data.
- **Removed Irrelevant Columns:** Dropped notes and unit columns as they didn't contribute to analysis.
- **Data Type Conversion:**
 - Converted the year column to numeric values and handled invalid entries
 - Cleaned the value column to remove non-numeric characters and convert to float
- **Standardized Text Columns:**
 - Converted type_of_connection to lowercase.
 - Trimmed spaces in service_provider and formatted month to title case.

```
Dataset Shape: (70728, 8)

Columns:
['year', 'month', 'circle', 'type_of_connection', 'service_provider', 'value', 'unit', 'notes']

Data Types:
year          int64
month         object
circle        object
type_of_connection object
service_provider object
value         object
unit          object
notes         object
dtype: object

Missing Values:
year          0
month         0
circle        0
type_of_connection 0
service_provider 0
value        12332
unit          0
notes        70217
dtype: int64
```

The image shows output of dataset and its shape, columns, data types and missing values in every column

Data Formatting for Analysis:

- Created Combined Date Column:
 - date = month + year → formatted using pd.to_datetime() for timeline visualizations.
- Separated Connection Types:
 - Created two DataFrames:
 - df_wireless → for mobile/wireless analysis
 - df_wireline → for fixed-line/wired connections
- Filtered Top Service Providers:
 - Extracted top 10 providers by frequency for focused visualization.

service_provider		count
BSNL		8650
Bharti Airtel		7391
Reliance Communications		5997
MTNL		5647
Vodafone		4064
Reliance Jio		3772
Vodafone Idea		3325
Tata Teleservices		2936
Idea		2682

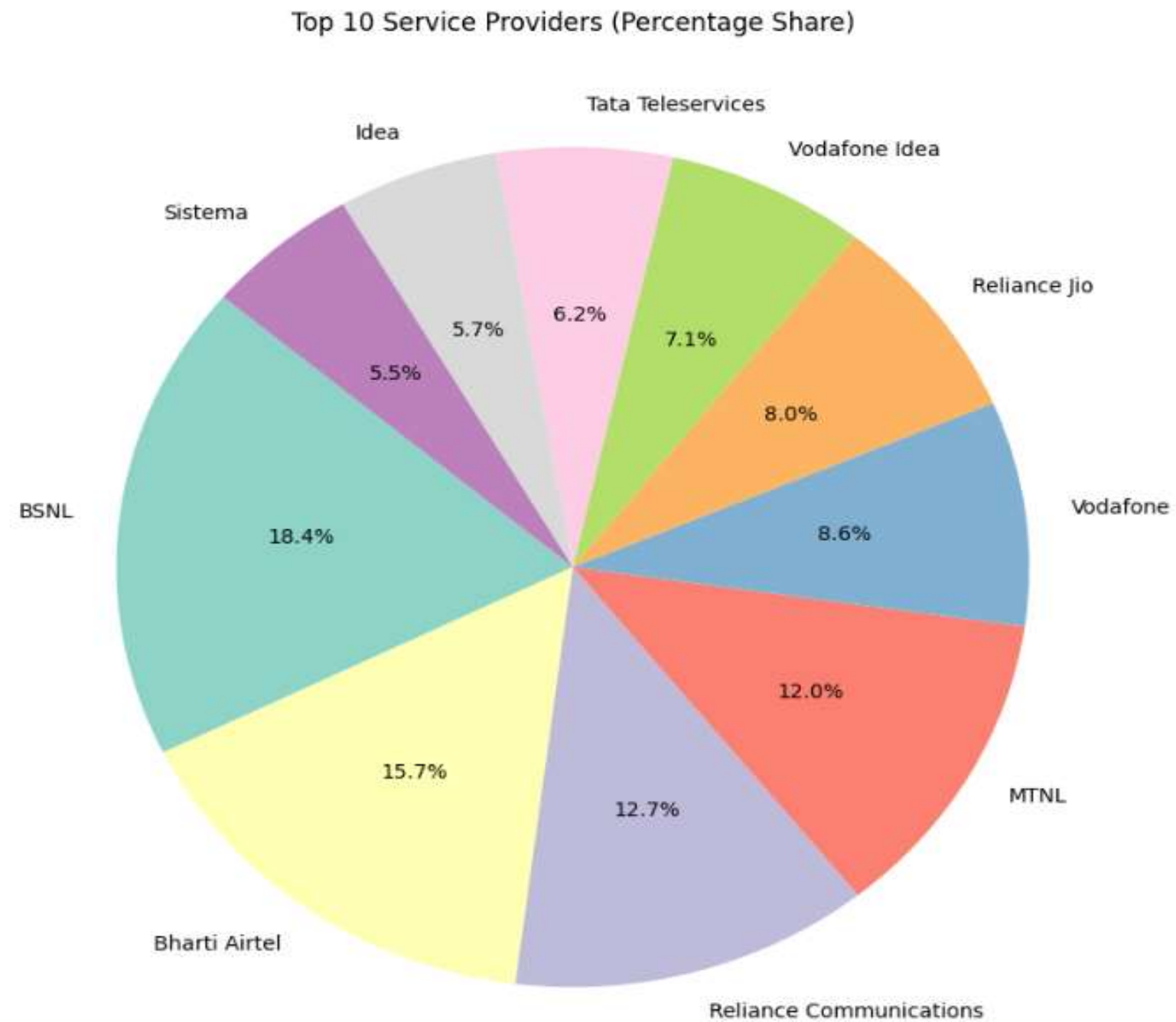
		0
year		int64
month		object
circle		object
type_of_connection		object
service_provider		object
value		float64
unit		object
notes		object

- Converted the year into numeric int and value into float



- These are some of the examples of service providers and their count.
- Checked provider distribution using df['service_provider'].value_counts().

Top Service Providers



Observation:

- The pie chart shows the distribution of telecom service providers based on their frequency in the dataset.
- BSNL has the largest share (18.4%), followed by Bharti Airtel (15.7%) and Reliance Communications (~12.7%).
- MTNL and Vodafone also have notable proportions, while providers like Sistema and Idea have smaller market shares.
- This indicates BSNL and Airtel are the most dominant service providers among users in the dataset.

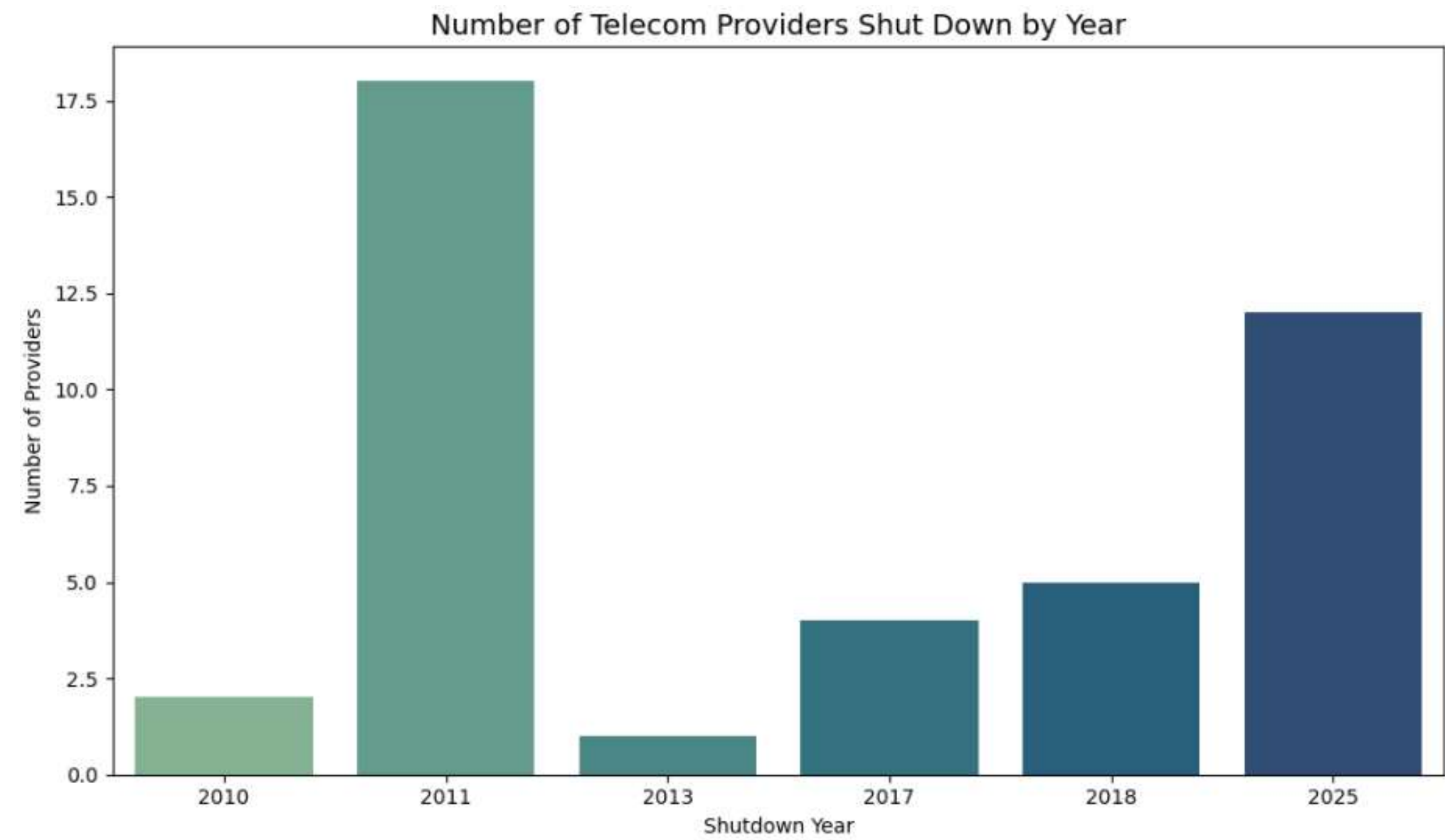
Wireless

- Conducted **trend analysis** on active and shutdown providers across years.
- Cleaned and standardized columns — removed unwanted symbols and converted data types.
- Beside are some of the examples showing Service provider and the month & year of shutting down the services.

	service_provider	shutdown_year	shutdown_month
0	Bharti Airtel (Including Tata Tele.)	2025	None
1	Reliance Communications	2017	September
2	Vodafone Idea	2025	None
3	BSNL	2025	None
4	MTNL	2025	None
5	Reliance Jio	2025	None

Key Insights:

- **BSNL** and **Airtel** remain dominant players with the highest subscriber shares.
- **Reliance Jio’s entry (post-2016)** triggered major competition and price disruption.
- **2011** recorded the highest number of shutdowns due to the **2G spectrum scam**.
- **2017–2018** saw consolidation — several smaller operators exited or merged.
- Market shifted from **20+ providers in 2010** to a few national players by 2025.
- **Jio, Airtel, and BSNL** continue to hold most of the wireless market share.



Wireline

- We made the same analysis as we did in the wireless data.

	year	month	circle	type_of_connection	service_provider	value	date
47465	2025	April	Andhra Pradesh	wireline	BSNL	836796.0	2025-04-01
47466	2025	April	Assam	wireline	BSNL	59049.0	2025-04-01

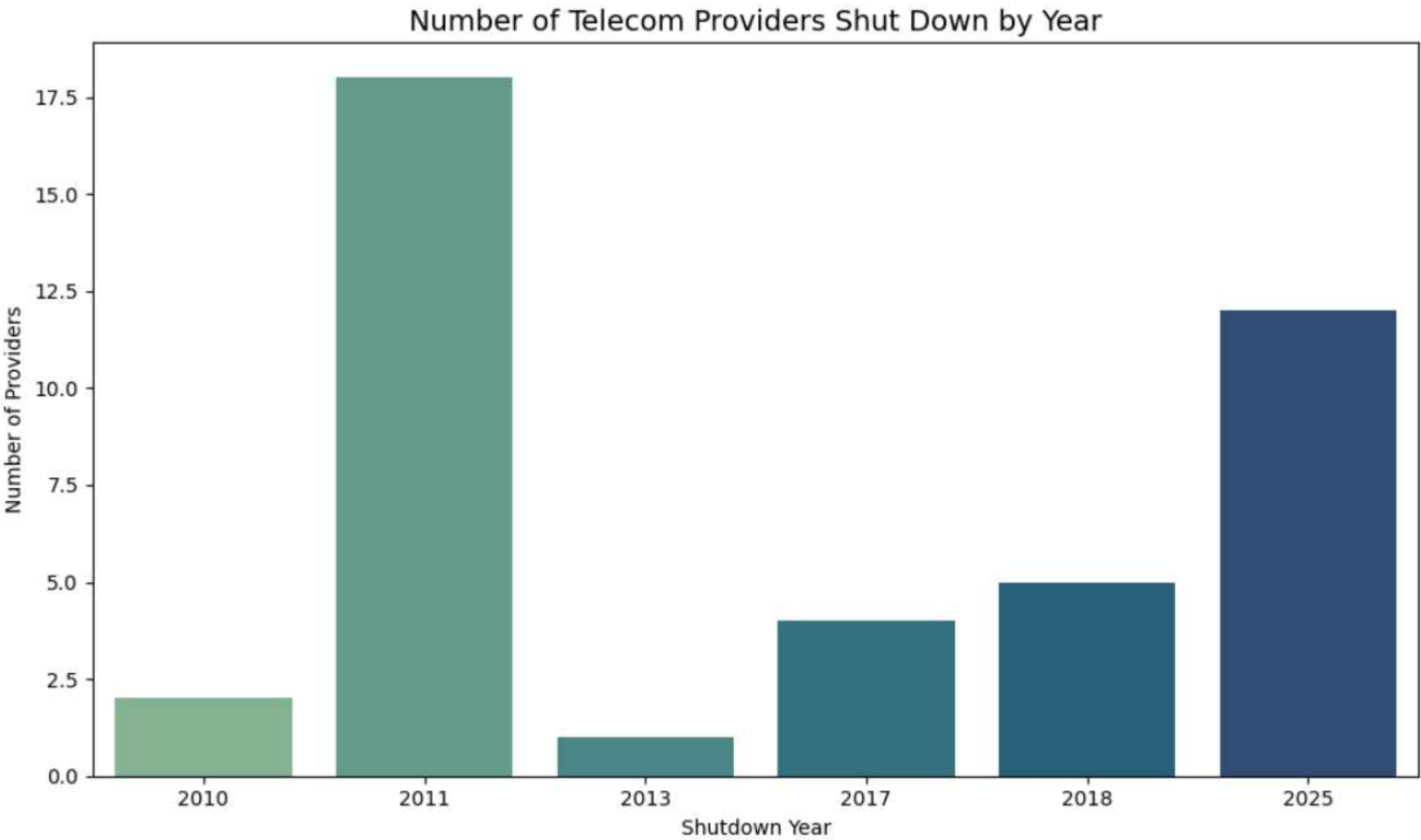
- These are couple of samples from the processed wireline data .
- Later the wireless and wireless data is merged for further analysis

Key Insights:

- The **wireline market contracted steadily** over time with few active providers remaining.
- 2011** again marked the peak of shutdowns following license cancellations.
- 2017–2018** experienced further decline as operators shifted focus to mobile services.
- By **2025**, only **BSNL, Airtel, and Jio** sustain wireline operations.
- The decline reflects **user migration to wireless and broadband alternatives**.
- BSNL** maintains presence due to government and rural connectivity programs.

	service_provider	shutdown_year	shutdown_month
0	Bharti Airtel (Including Tata Tele.)	2025	None
1	Reliance Communications	2017	September
2	Vodafone Idea	2025	None
3	BSNL	2025	None

Service providers that are shut down along with year & month of shutting down



Feature Engineering

Feature Construction & Derived Metrics

- **Temporal Features:**
 - Created lag-based variables to capture time dynamics in each operator–circle pair. These lags help model *momentum, trends, and seasonality* in subscriber base or revenue metrics.
- **Date Components:**
 - year and month to enable grouping and time-series visualization.
- **Circle Type:**
 - Classified circles into **Metro** and **Non-Metro** categories. Enables differentiation of market maturity and customer churn patterns.
- **Connection Type:**
 - Binary encoded type of connection: `is_wireless` = 1 if “wireless”, else 0. Provides separation between broadband and mobile markets.
- **Operator Geographic Diversity:**
 - Measured how widely each operator is present, Indicates national reach and resilience to localized churn.
- **Market Size Category:**
 - Ranked circles based on their cumulative market volume, Helps distinguish between large and small telecom markets.
- **Competitive & Market Dynamics Features:**
 - `market_share` = share of each operator’s value within its circle per month,
 - `market_rank` = rank of operator based on share in that circle and month. Provides relative standing among competitors.

Derived Market Indicators

Herfindahl–Hirschman Index (HHI)

- Measures market concentration:
 - High HHI → few dominant players (less competitive).
 - Low HHI → more fragmented market (high competition).
 - Used to assess circle-level competitiveness.

Market Composition Metrics

- Shows how market share is distributed among operators and circles.
Helps identify dominant players, competitive intensity, and regions with potential churn risk or market opportunity.

Volatility and Stability Indicators

- Measure the consistency of market share and rank movements over time.
Higher volatility indicates unstable markets with high churn potential and shifting competitive positions.

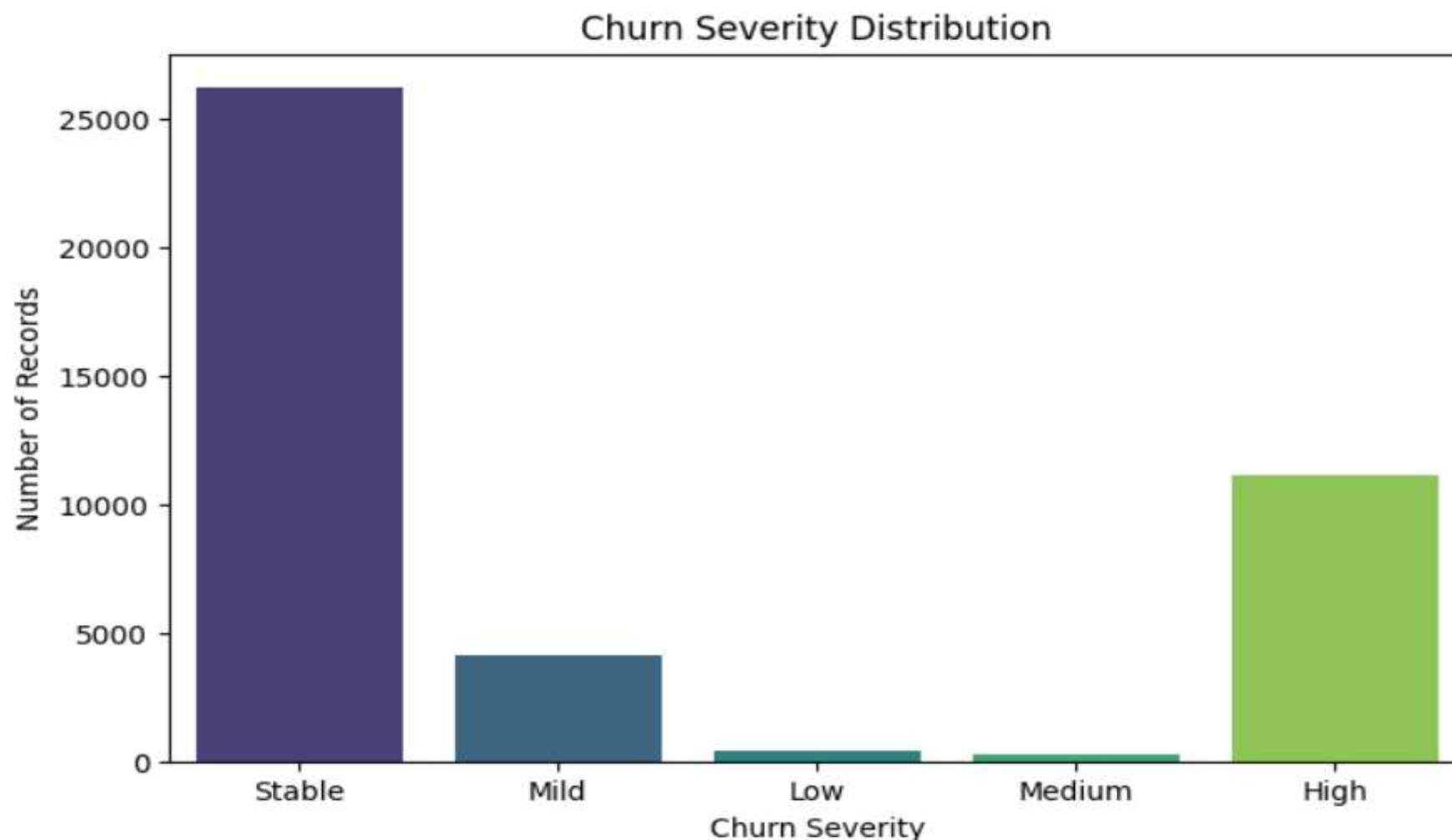
Feature Aggregations

- Aggregate and transform existing variables to extract meaningful trends.
Examples include lag ratios or combined indices (HHI + churn) to classify markets as stable, competitive, or volatile.

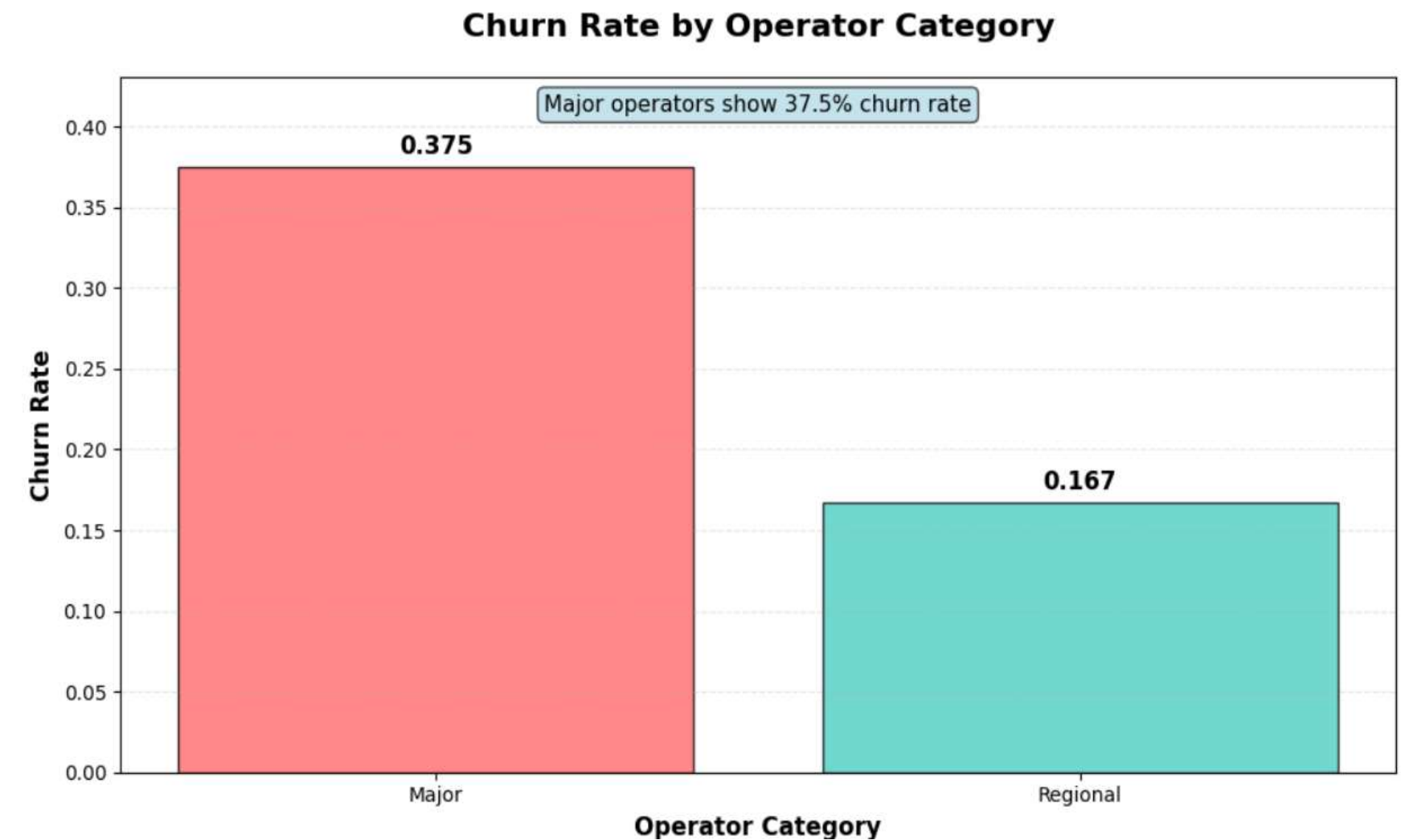
Churn Severity

- Quantifies market instability:
 - High values indicate rapid loss/gain of market share.
 - Aggregated at operator-circle level to prioritize high-risk zones.

- Churn Categories

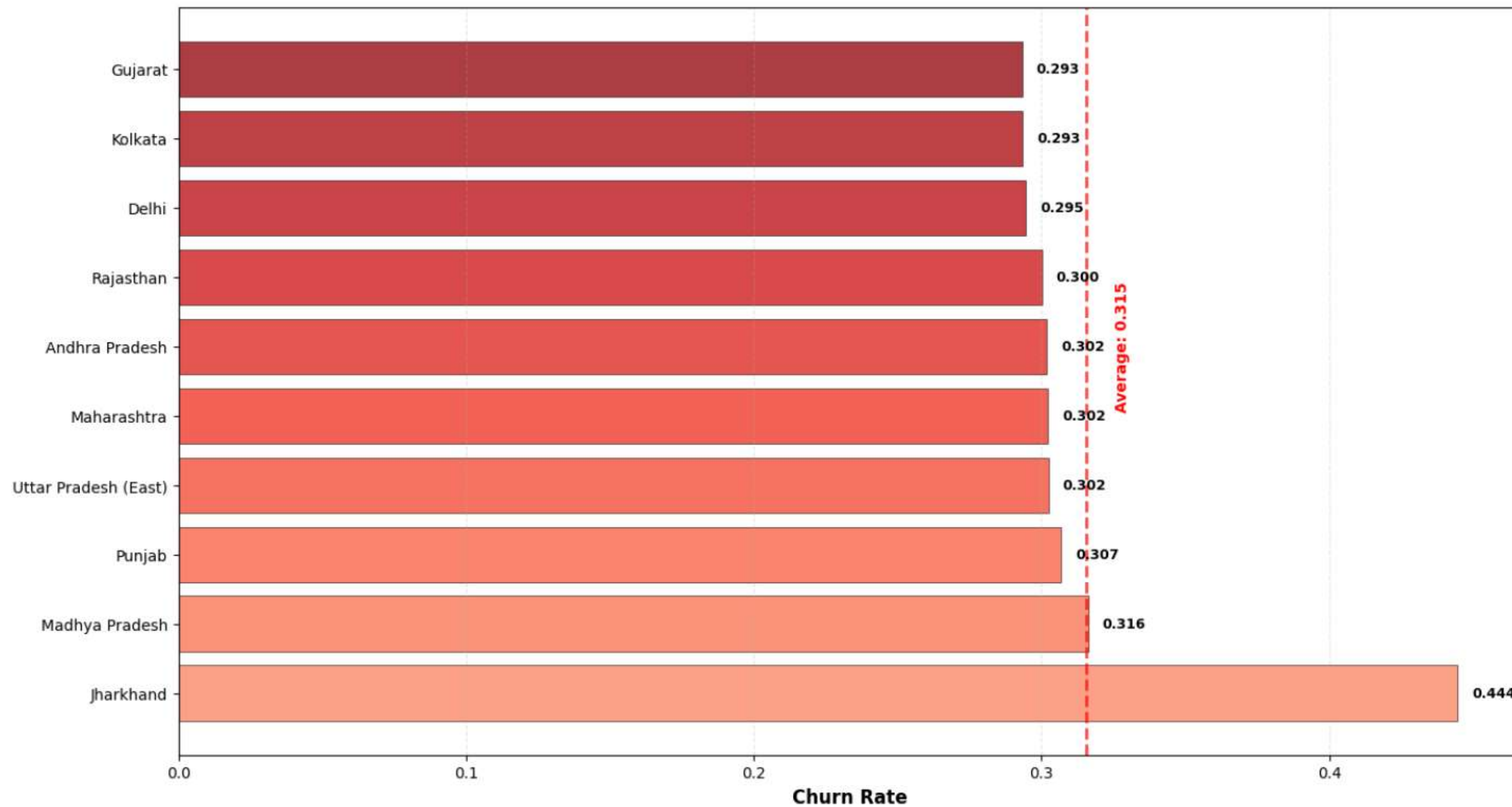


- The majority (26,205) are *Stable*, which is good for the company.
- But 11,110 customers ($\approx 28\%$) are *High churn risk*, which is significant and needs attention.



- Major operators are experiencing a significantly higher churn rate compared to regional ones.
- Despite strong brand presence, about 1 in 3 customers of major players are discontinuing service month-over-month.

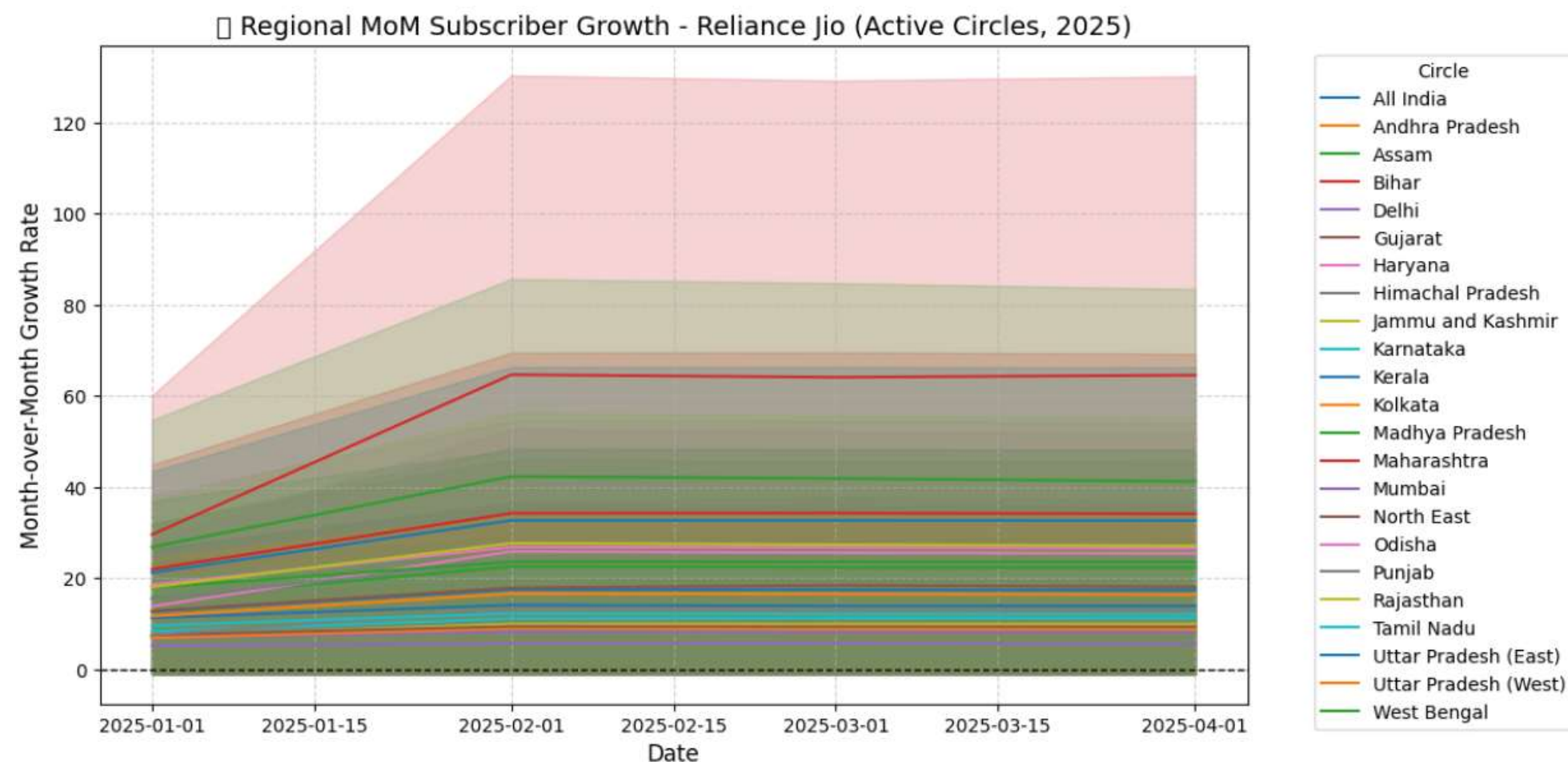
Top 15 Circles by Churn Rate



Top 10 Circles Churn Analysis:

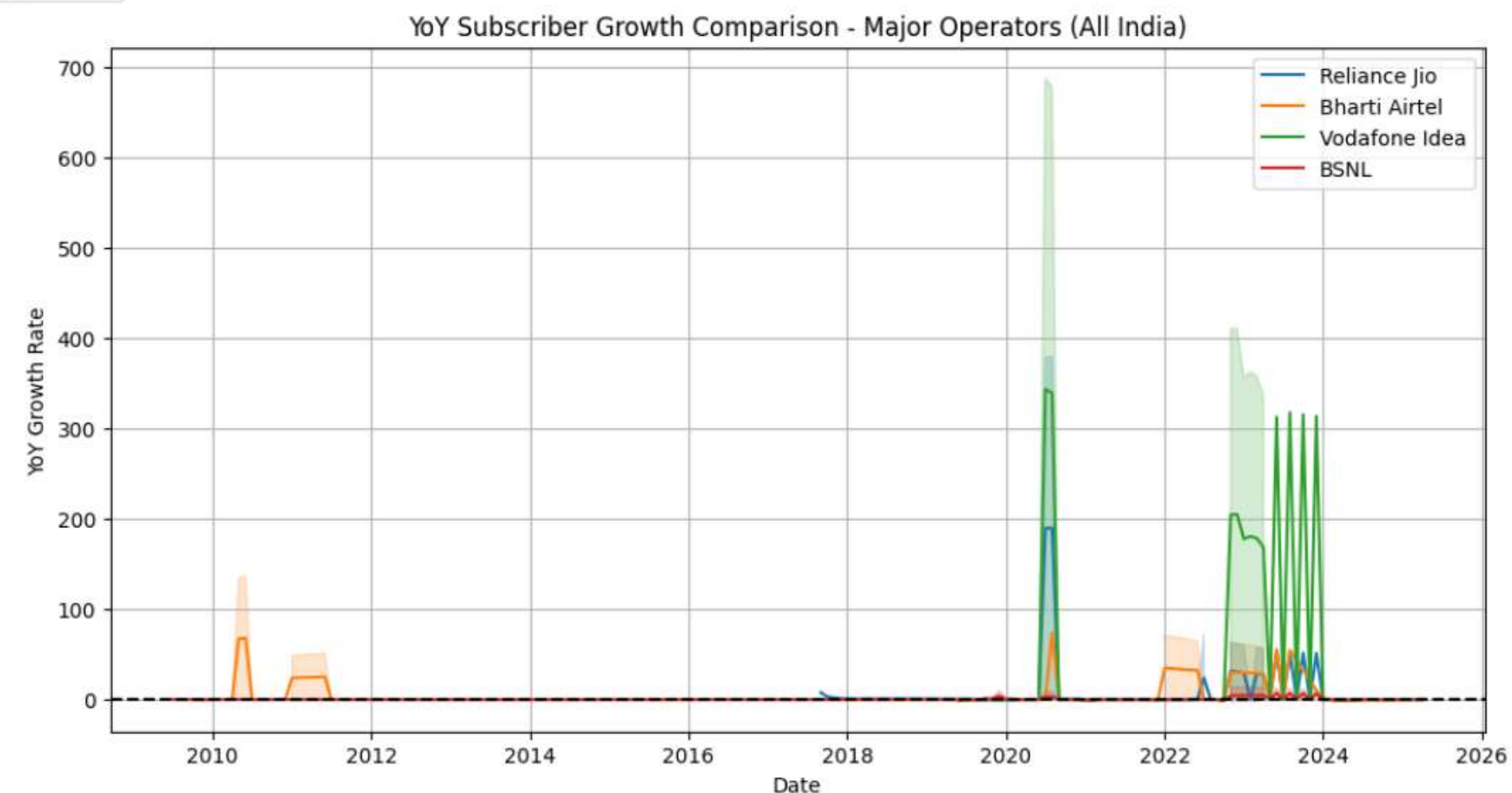
- Highest Churn: Jharkhand (0.444)
- Average Churn (Top 10): 0.315
- Range: 0.293 - 0.444

- Growth Comparisons

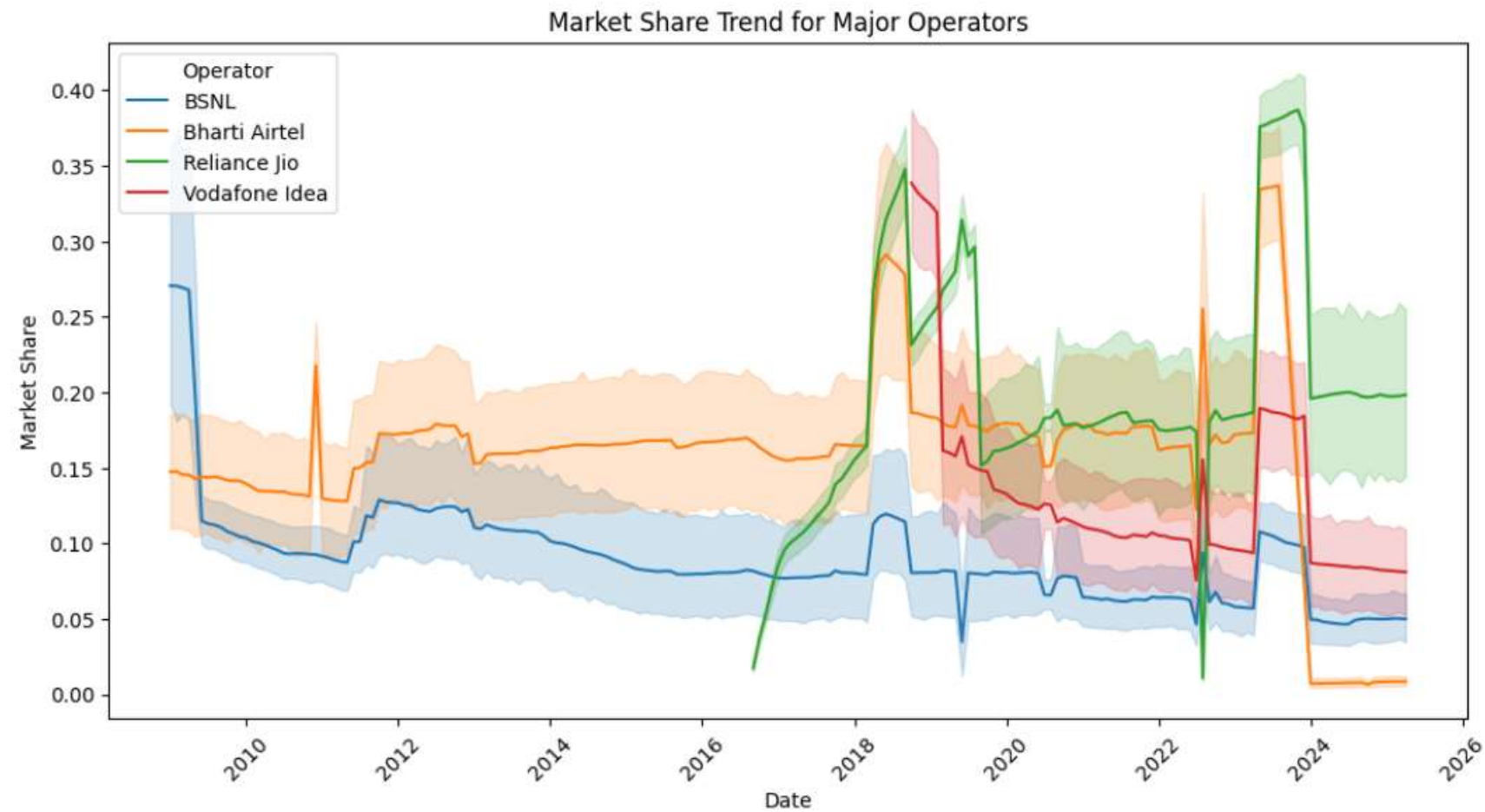


- Shows **month-over-month subscriber growth** across different telecom circles in India.
- Indicates that **Bihar, Maharashtra, and All-India** circles have the highest growth spikes.
- The chart highlights **regional variations**, showing stronger growth in some circles compared to others

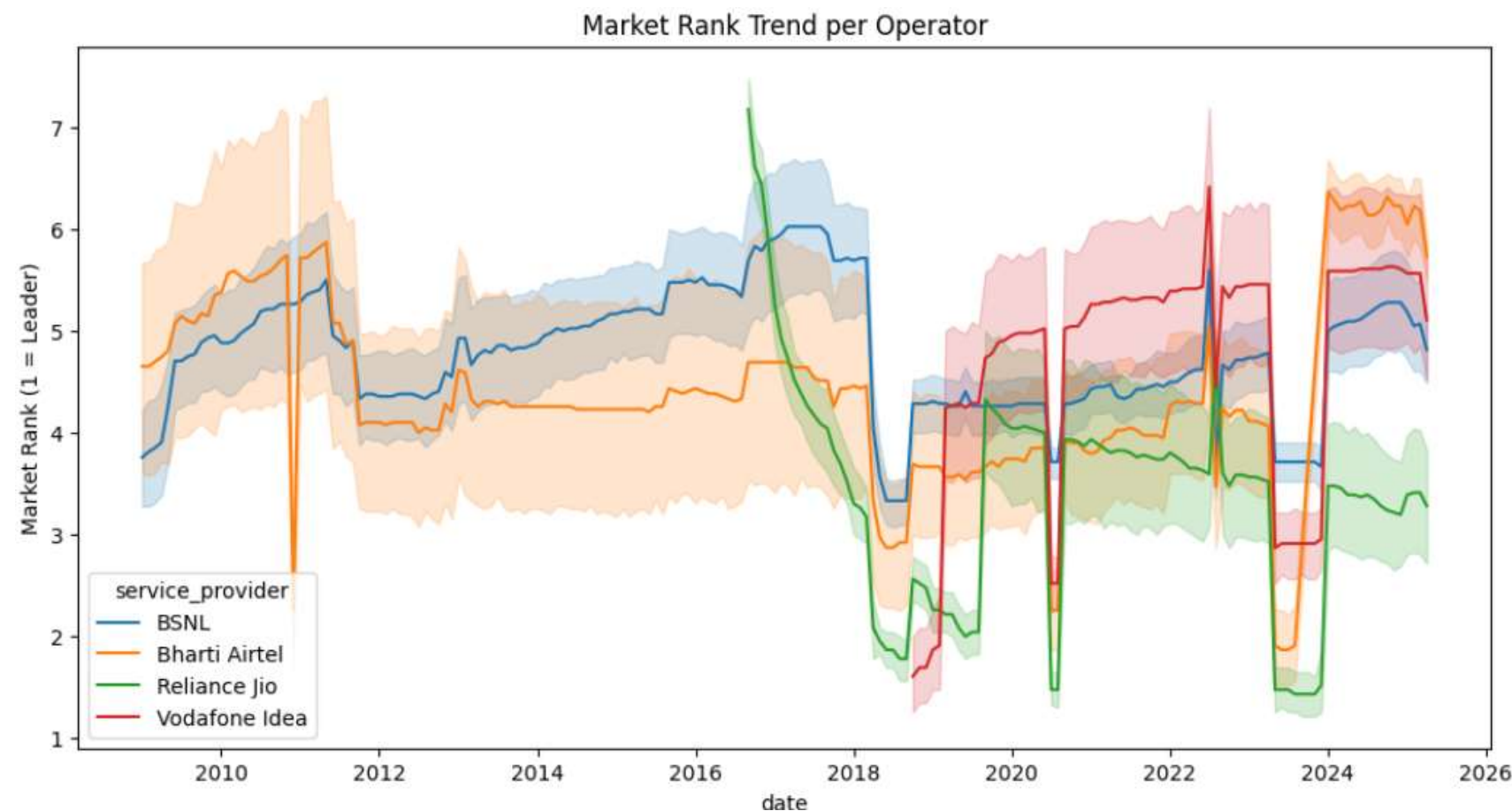
- Compares **year-over-year growth rates** of major telecom operators — Jio, Airtel, Vodafone Idea, and BSNL.
- Reliance Jio and Vodafone Idea show **sharp spikes around 2020**, indicating aggressive expansion or subscriber gains.
- Airtel shows moderate but **stable growth**, while BSNL remains comparatively flat.



- Market Trends

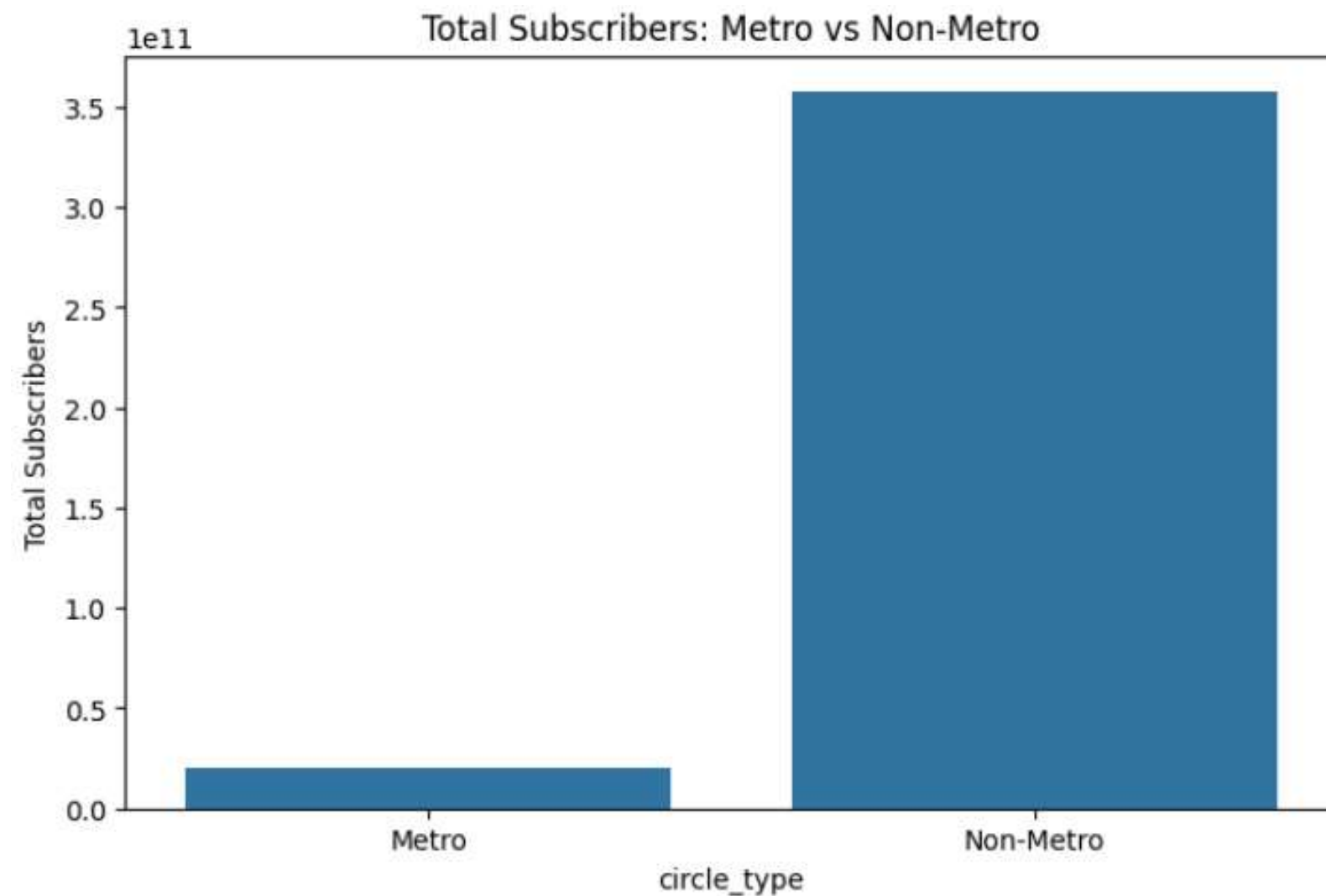


- Reliance Jio (green) rises sharply around 2016–2018, overtaking competitors.
- Bharti Airtel and Vodafone Idea fluctuate but remain relatively stable before Jio's entry.
- BSNL shows a steady decline in market share over the years.
- Operators losing market share consistently (BSNL, Vodafone Idea at times) are more likely to experience subscriber churn.

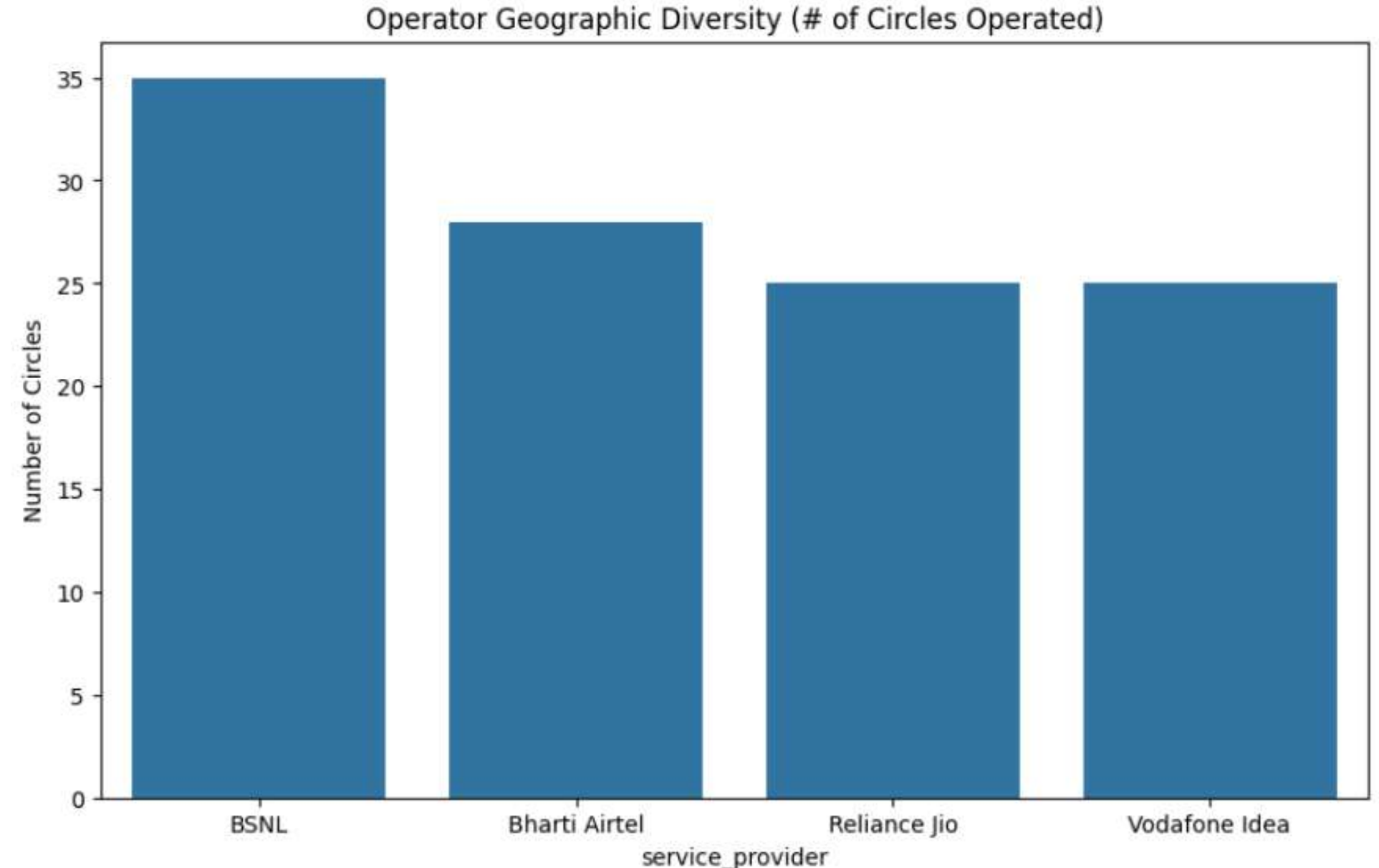


- Tracks the **market position (1 = leader)** of each telecom provider over time.
- Reliance Jio rapidly climbed to the **top rank after 2016**, overtaking others.
- Vodafone Idea's rank shows a **declining trend**, while Airtel maintains **steady competitiveness**.
- BSNL has remained **lower-ranked** throughout the observed period.

- Subscribers and Operator Diversity



- Compares total subscriber counts between **Metro** and **Non-Metro** circles.
- Non-Metro regions dominate with **a significantly larger subscriber base**.
- Highlights the **strong rural and semi-urban market presence** for telecom operators.



- BSNL covers 35 circles, which is higher than all other operators.
- Bharti Airtel operates in 28 circles, slightly fewer than BSNL
- Operators with greater geographic diversity (BSNL, Airtel) potentially have a larger market reach and can capture more subscribers.
- Operators with fewer circles (Jio, Vodafone Idea) may focus on densely populated or profitable circles rather than blanket coverage.

Model Deployment and Validation

Models Used :

- Random Forest
- XG Boost
- Light GBM
- Cat Boost
- Artificial Neural Network (ANN)

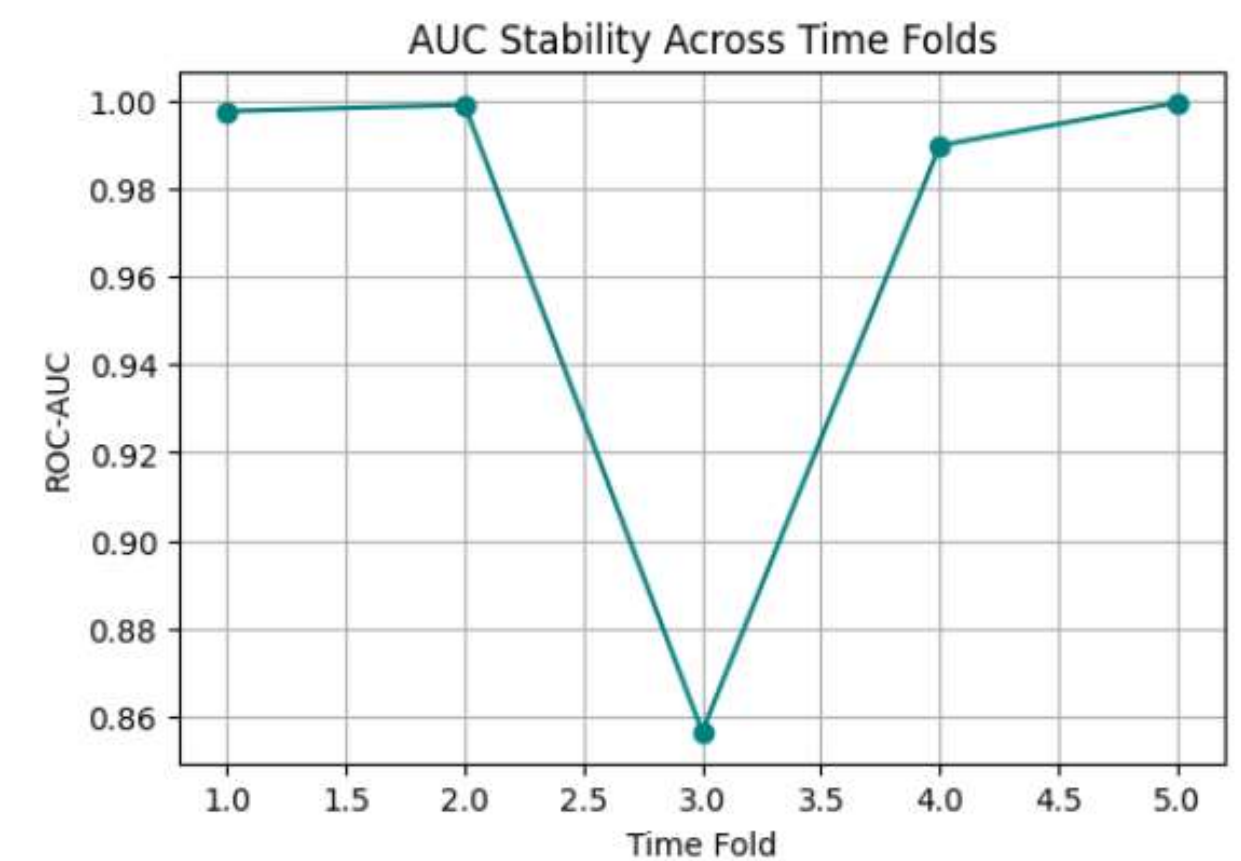
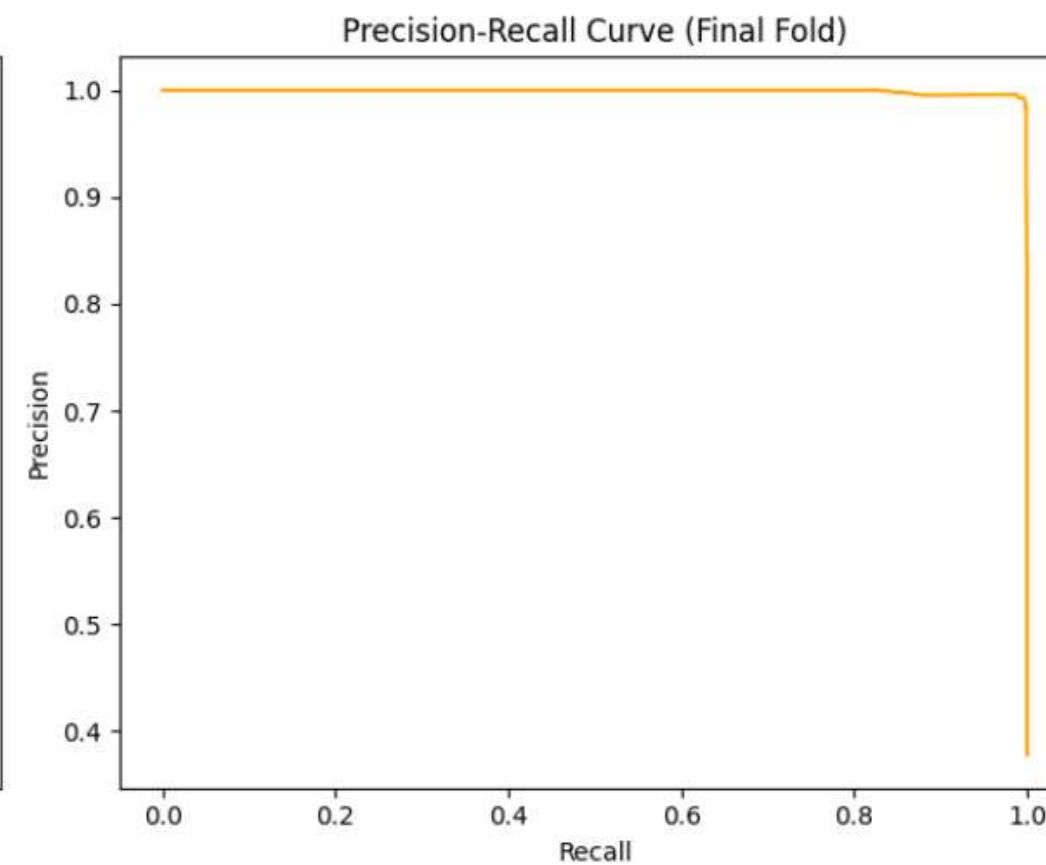
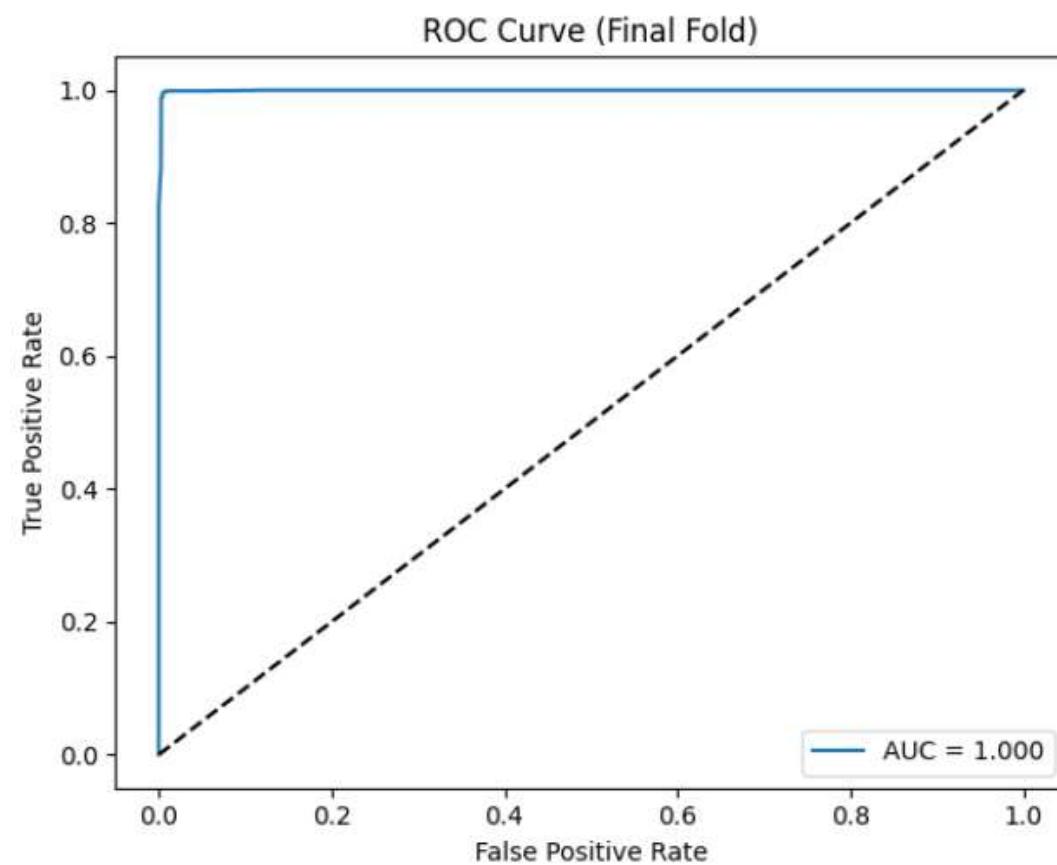
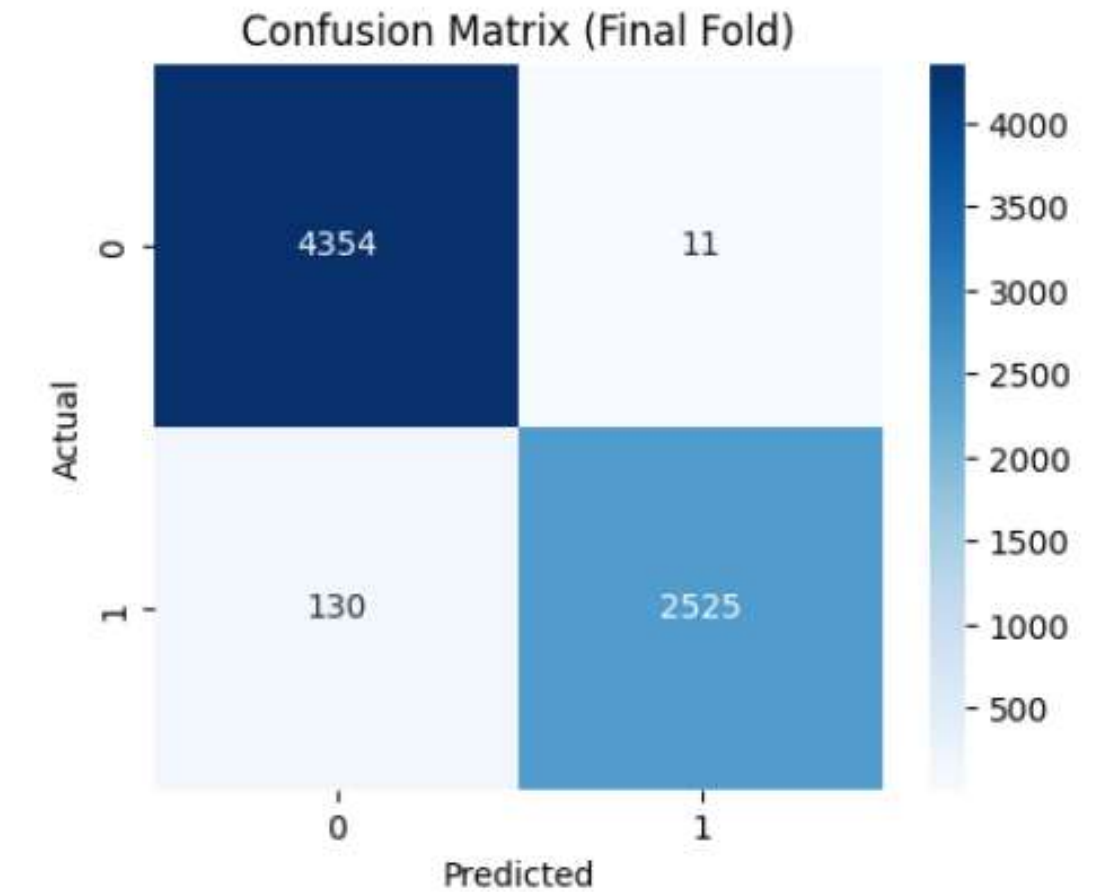
Cross-Validation Setup :

- Used Time Series Split (5 folds)
- Metrics: Accuracy, F1-score, ROC-AUC.
- Purpose: to ensure robustness and avoid data leakage.

Random Forest :

Metrics:

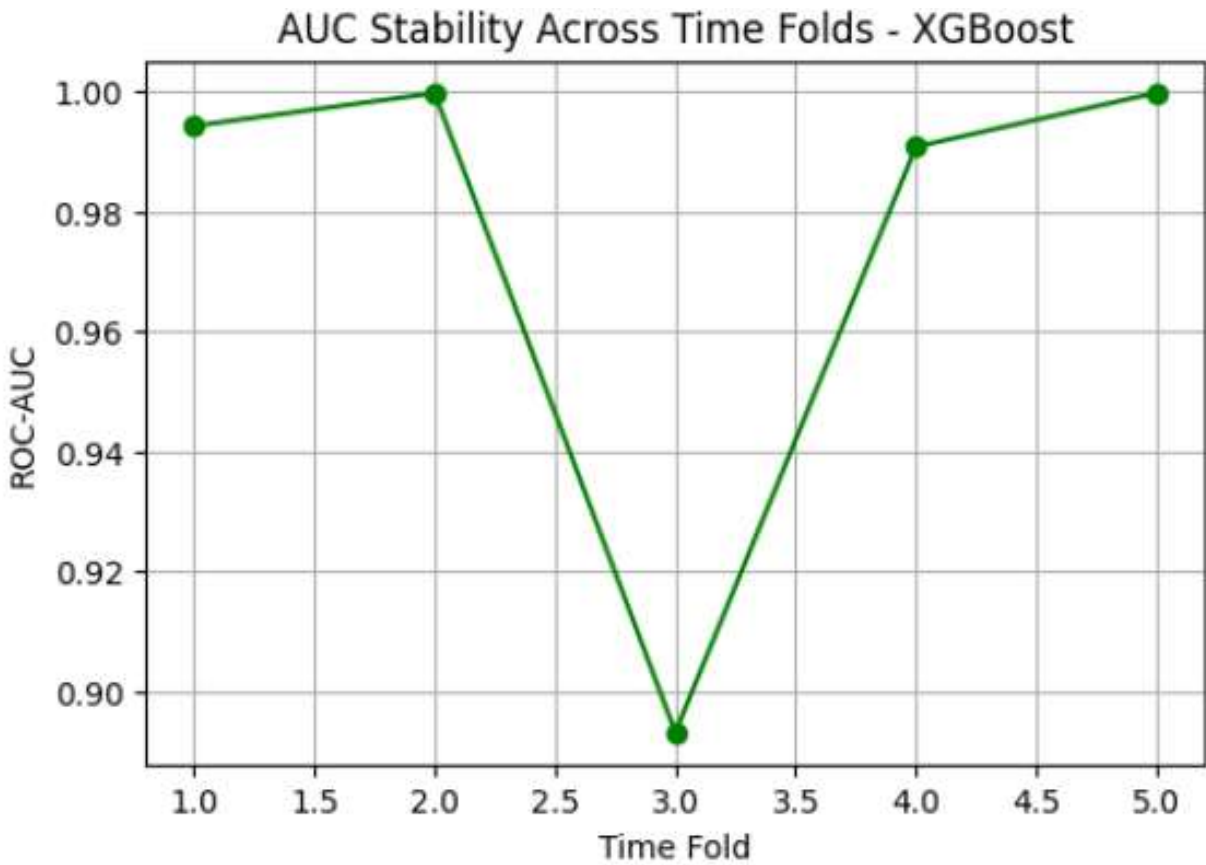
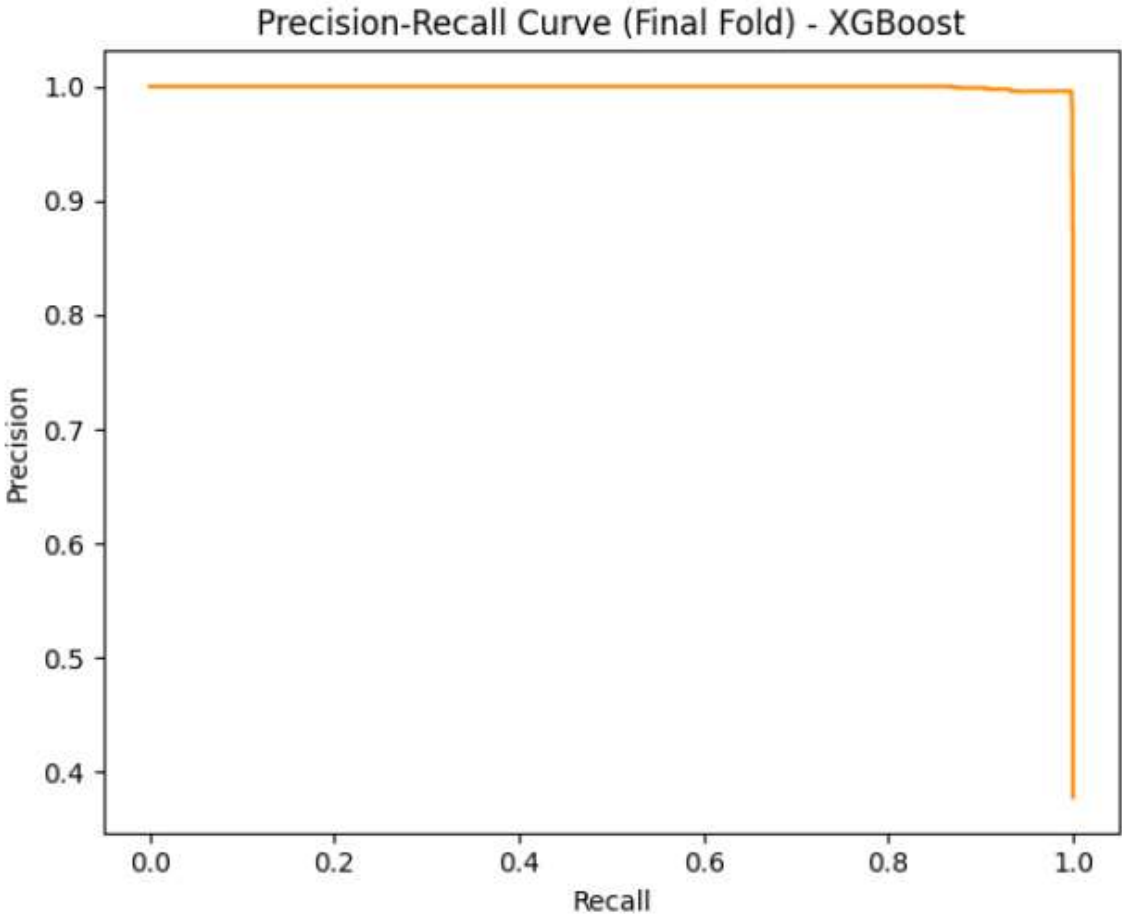
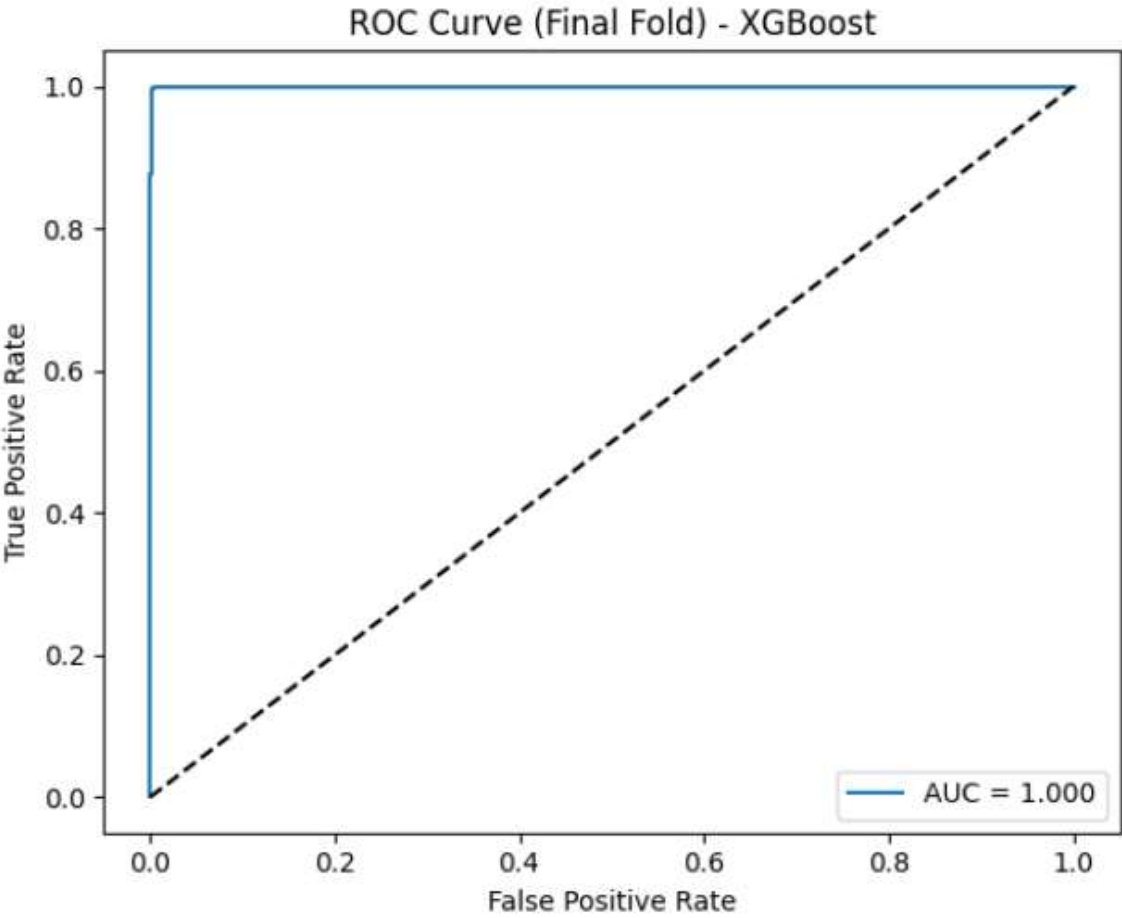
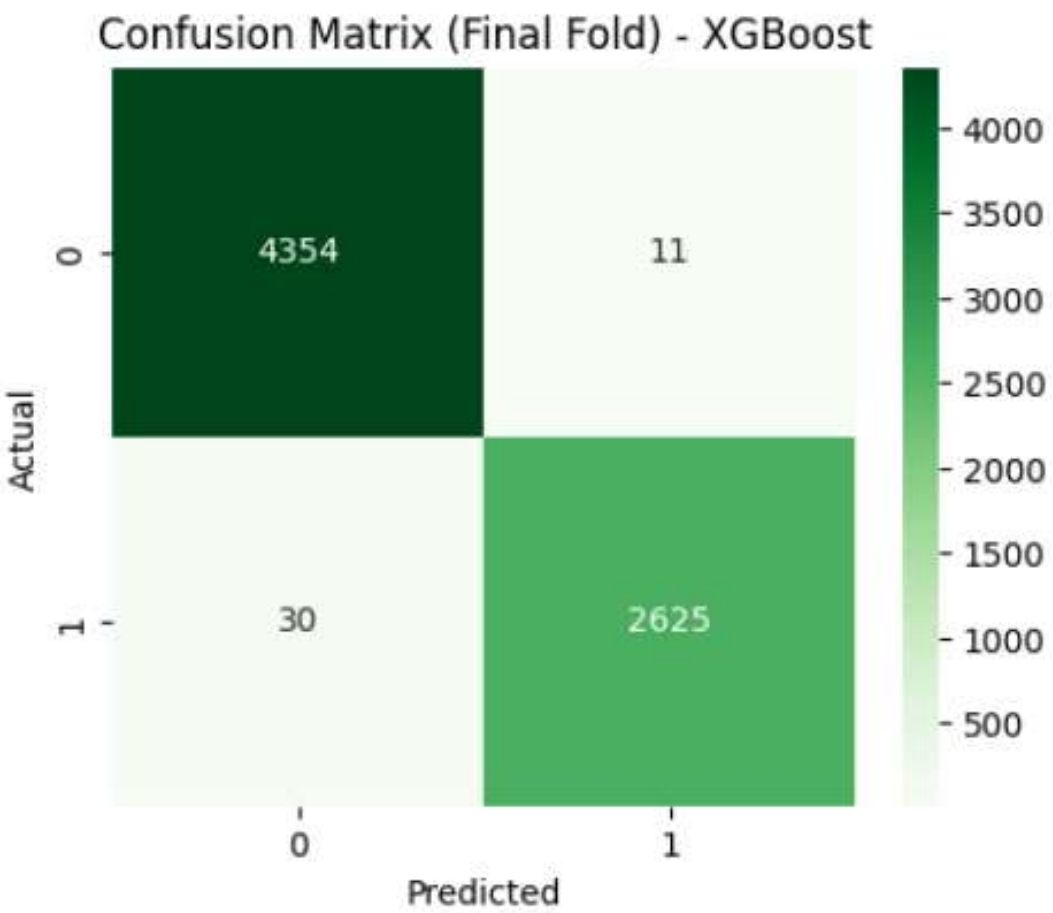
- Accuracy: **0.9679**
- F1-score: **0.9158**
- ROC-AUC: **0.9685**
- Precision: **0.9158**
- Recall: **0.9164**
- Precision@Top10%: **0.9536**
- The Random Forest model achieved **high overall accuracy (96.8%)** with balanced precision and recall (~91%).
- The **confusion matrix** shows very few misclassifications, confirming strong reliability.
- The **ROC curve (AUC = 1.0)** and **Precision-Recall curve** indicate near-perfect discrimination between classes.
- **AUC stability across folds** proves the model's consistency and robustness over time.
- Overall, Random Forest demonstrates **excellent predictive performance** and serves as a **reliable baseline model**.



XG Boost :

Metrics:

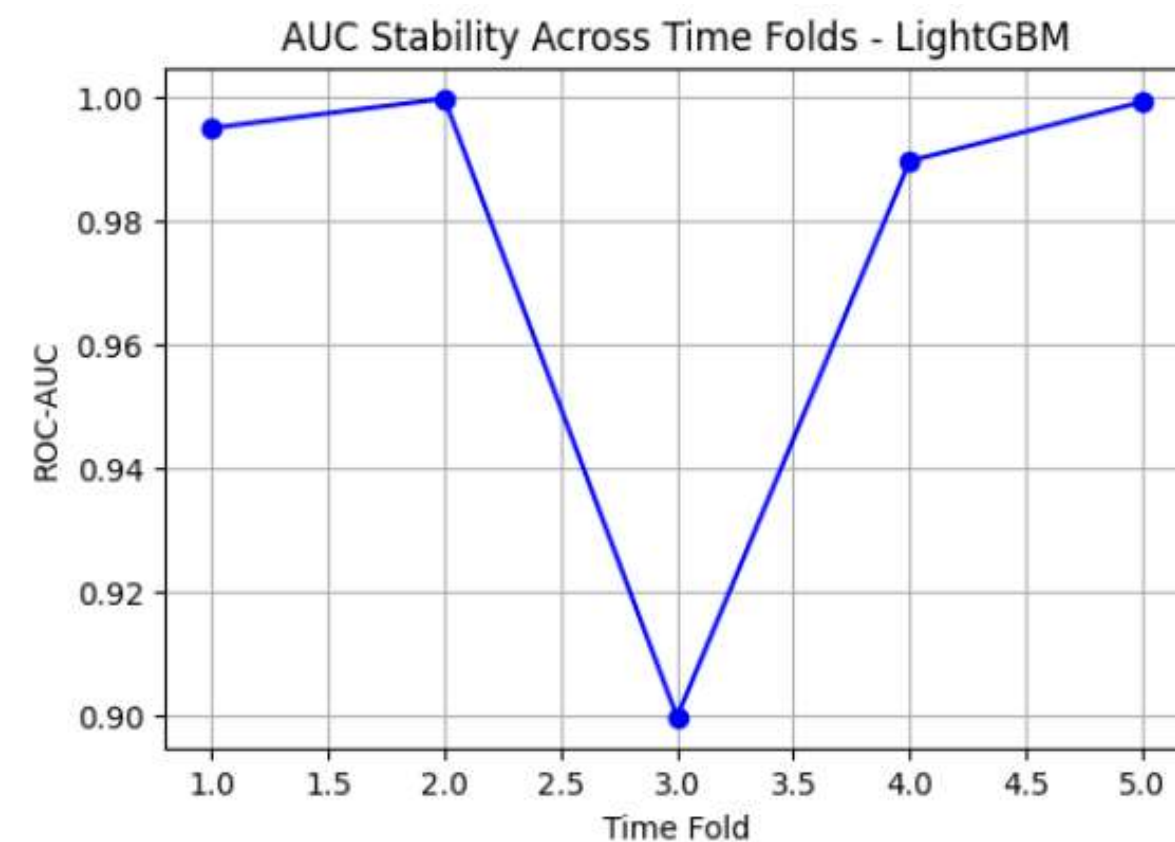
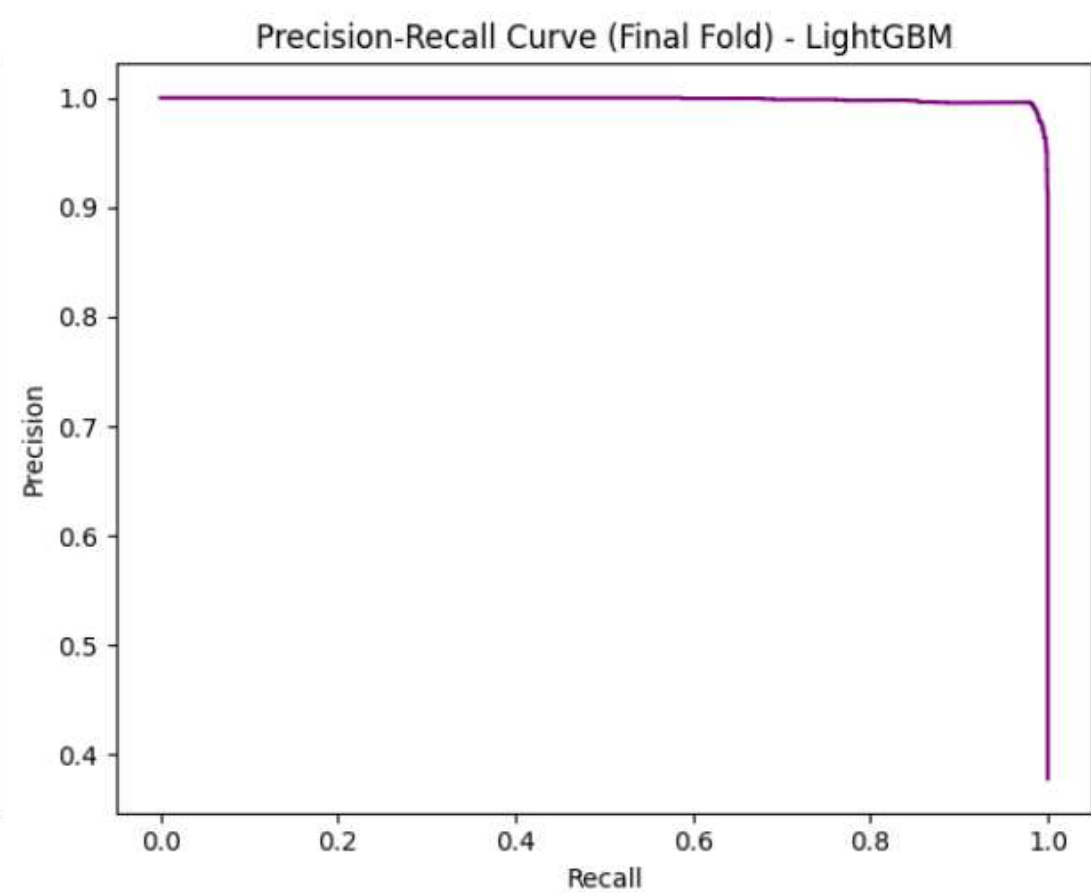
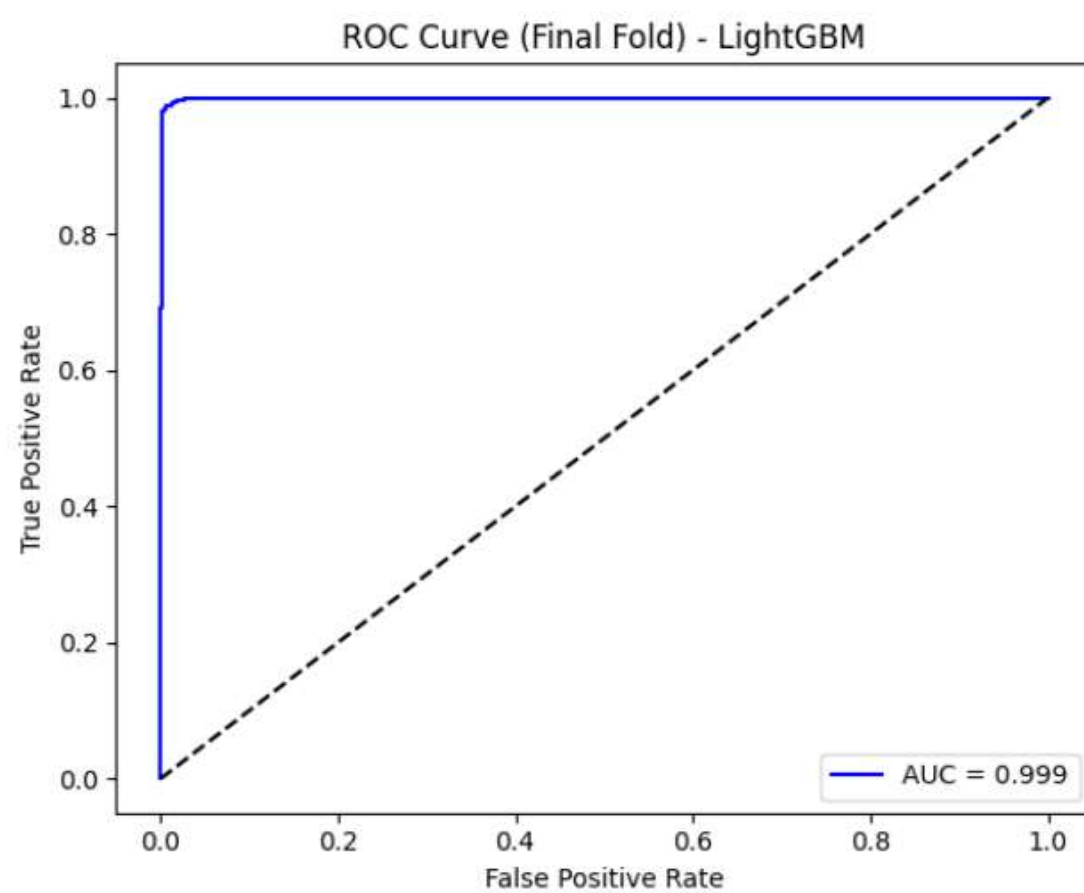
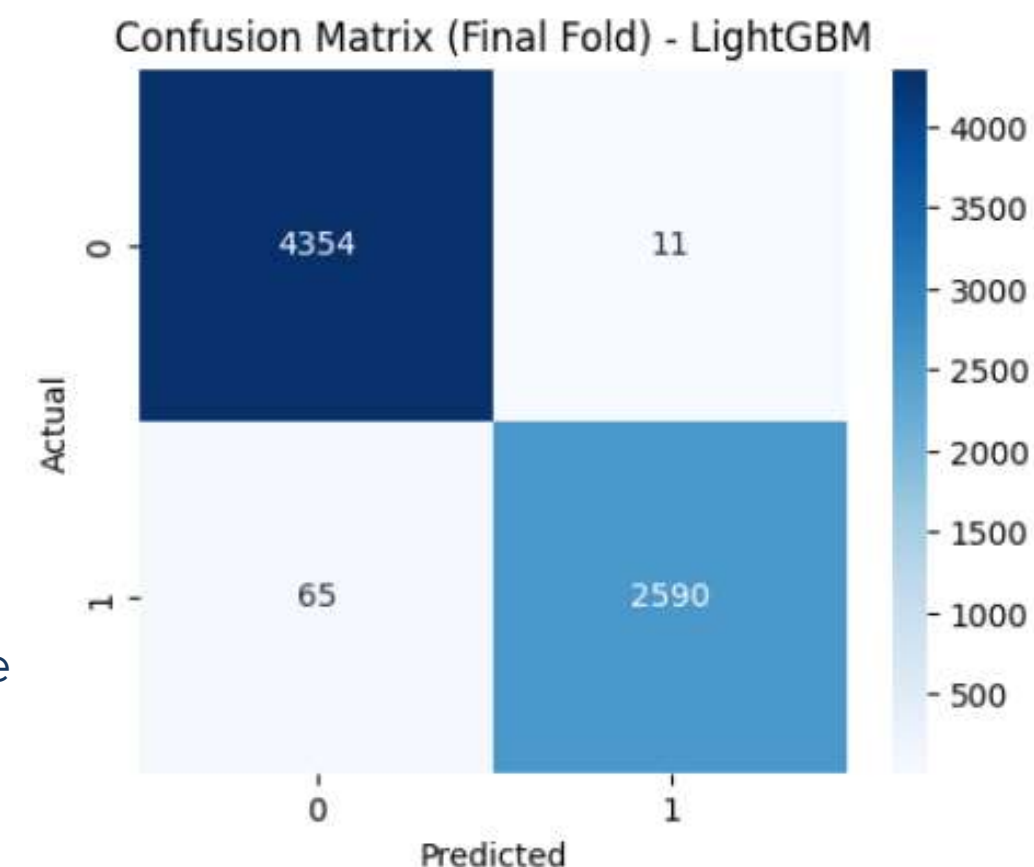
- Accuracy : **0.9675**
- F1-score : **0.9135**
- ROC-AUC : **0.9755**
- Precision : **0.9005**
- Recall : **0.9275**
- Precision@Top10% : **0.9963**
- XGBoost achieved **excellent accuracy (96.7%)** with a strong F1-score (0.9135) and high recall (0.9275).
- The **confusion matrix** shows very few false predictions, proving the model's reliability.
- **ROC Curve (AUC = 1.0)** and **Precision-Recall curve** reflect near-perfect classification capability.
- **AUC stability** indicates consistent performance across time folds with minor variation in one fold.



LightGBM :

Metrics :

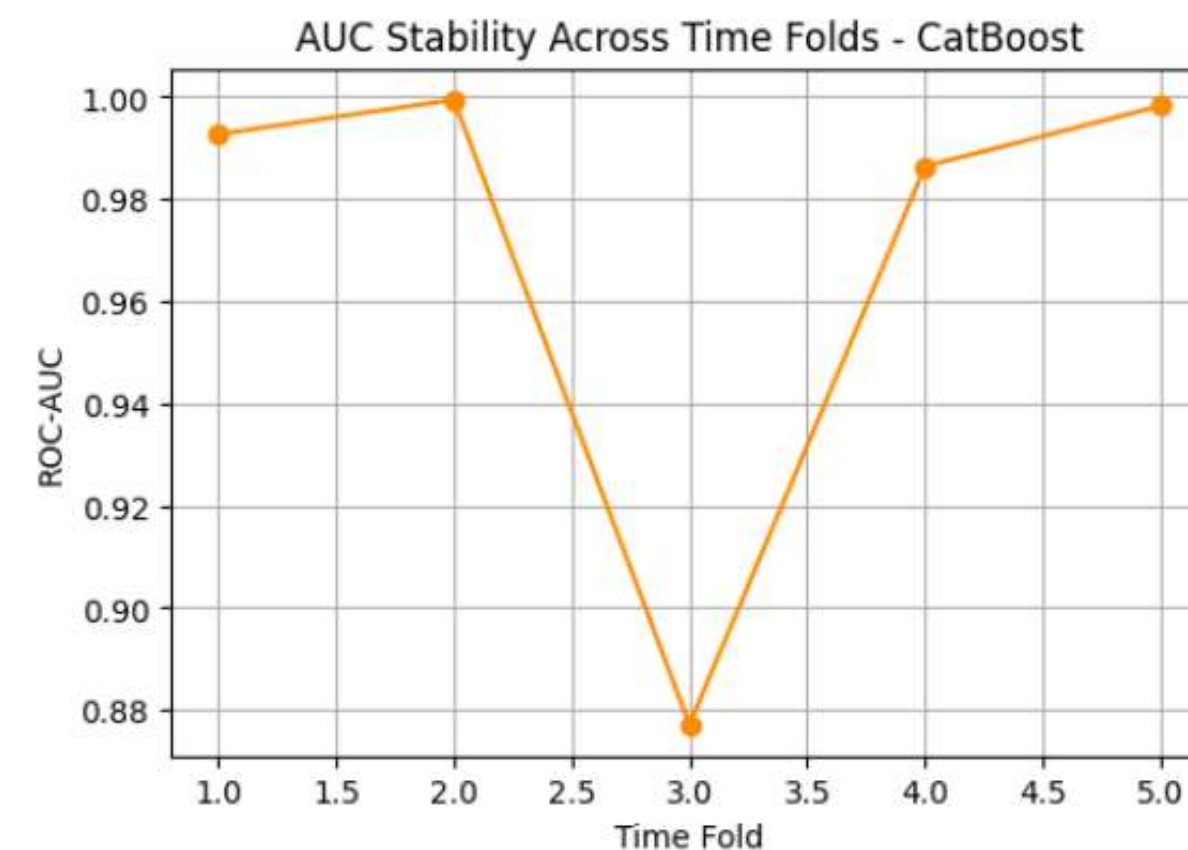
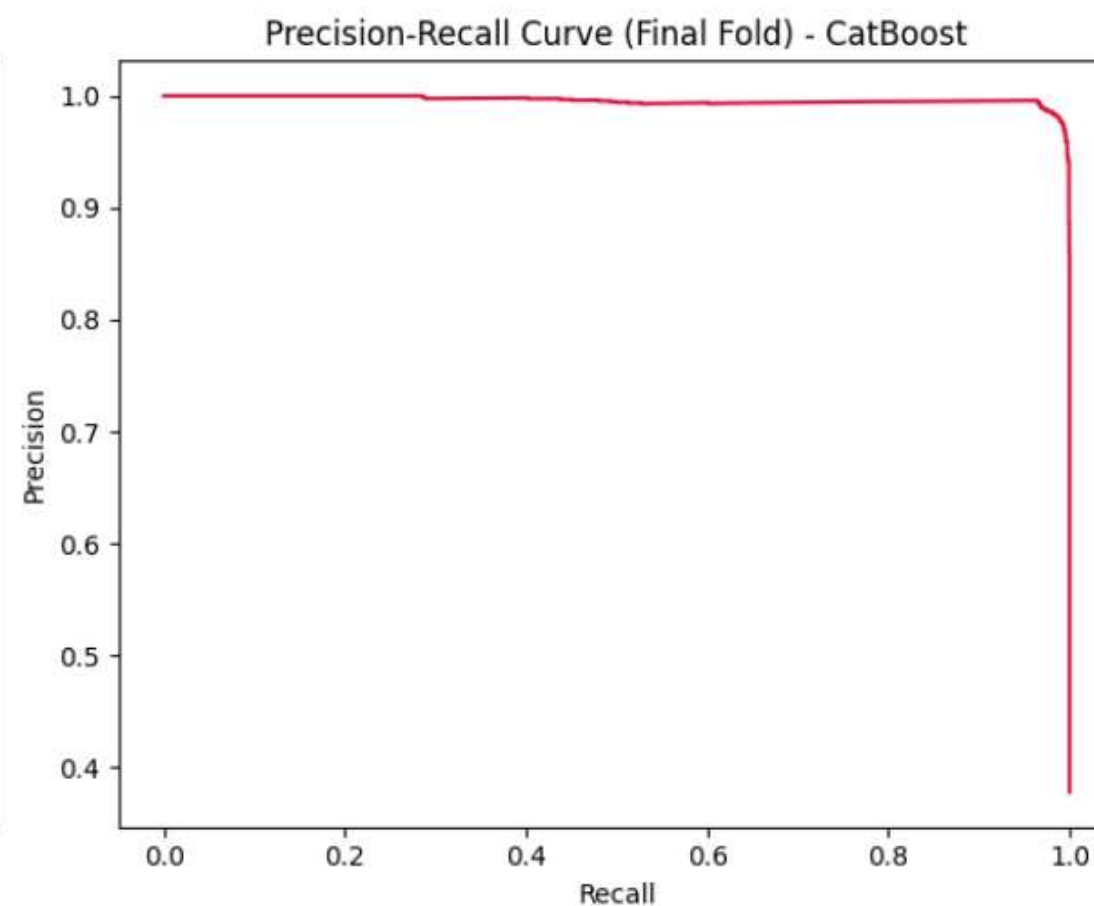
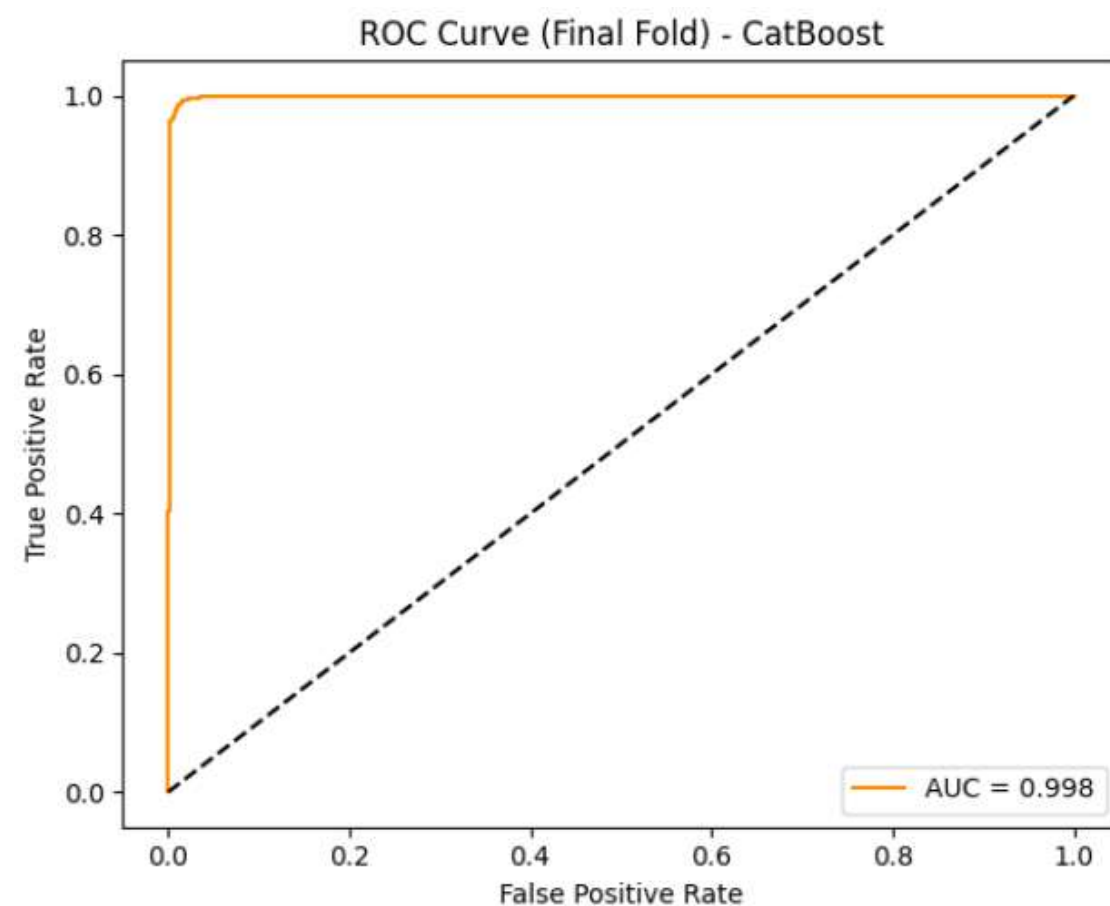
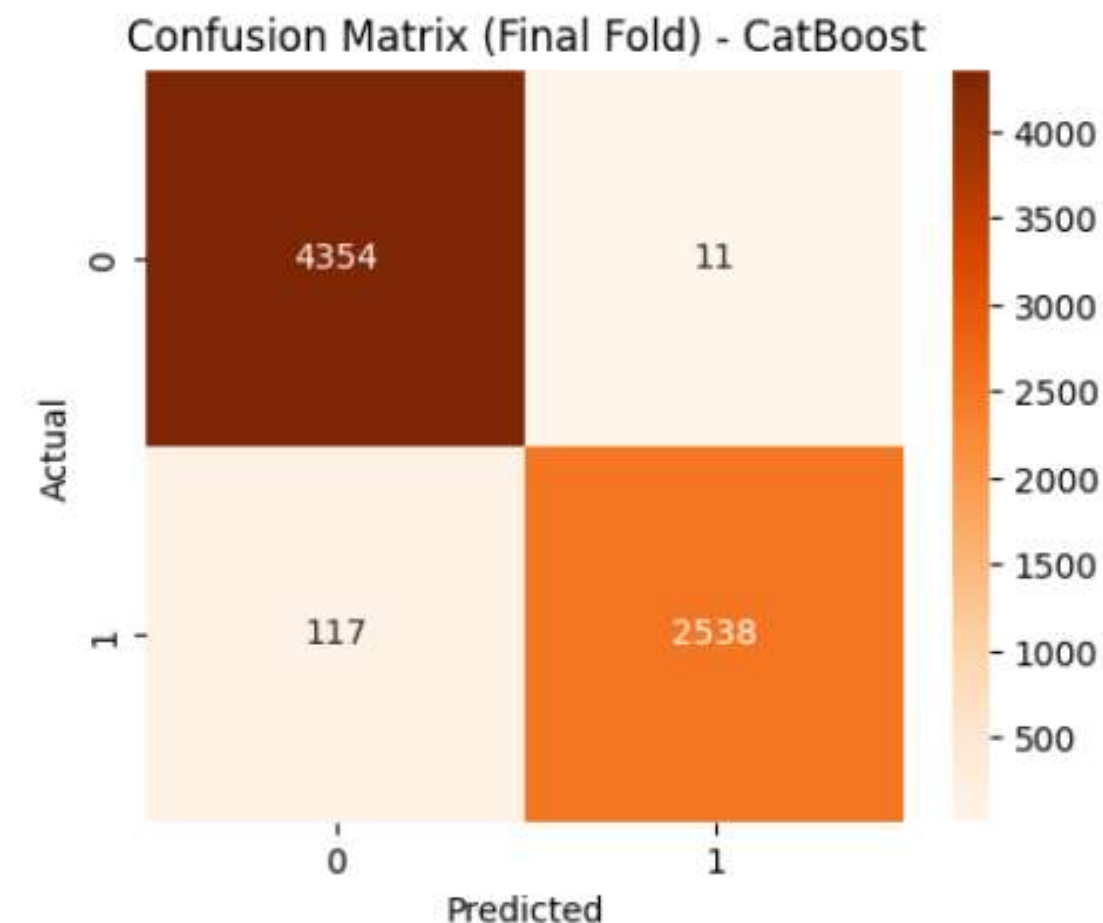
- Accuracy : **0.9721**
- F1-score : **0.9251**
- ROC-AUC : **0.9767**
- Precision : **0.9250**
- Recall : **0.9258**
- Precision@Top10% : **0.9823**
- LightGBM delivered **excellent predictive performance** with an overall accuracy of **97.2%** and a strong **F1-score of 0.9251**.
- The **ROC-AUC of 0.9767** indicates outstanding model discrimination between classes.
- The **Precision (0.9250)** and **Recall (0.9258)** show a well-balanced detection of both positive and negative cases.
- The **Precision@Top10% of 0.9823** demonstrates high reliability in top predictions.
- Performance remained **stable across time folds**, confirming LightGBM's robustness and consistency over temporal data.



CatBoost :

Metrics

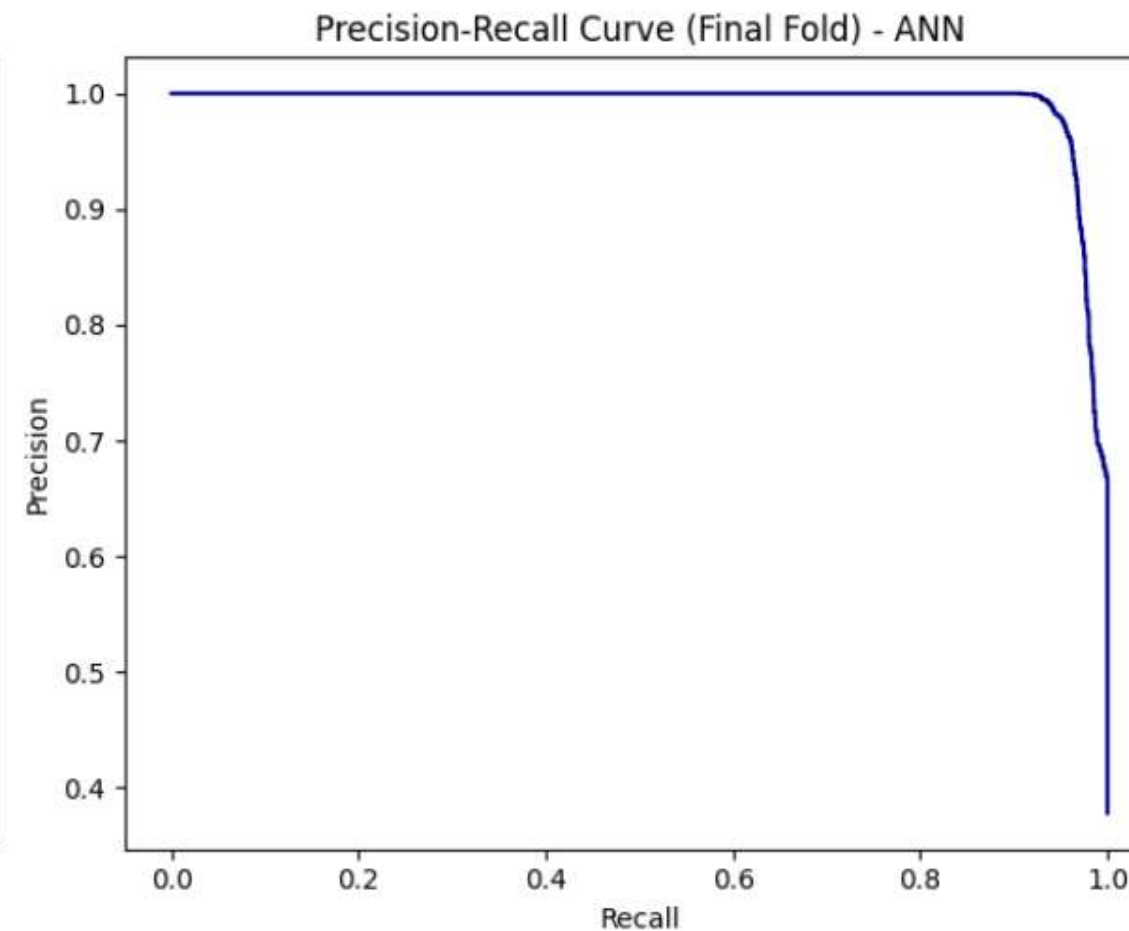
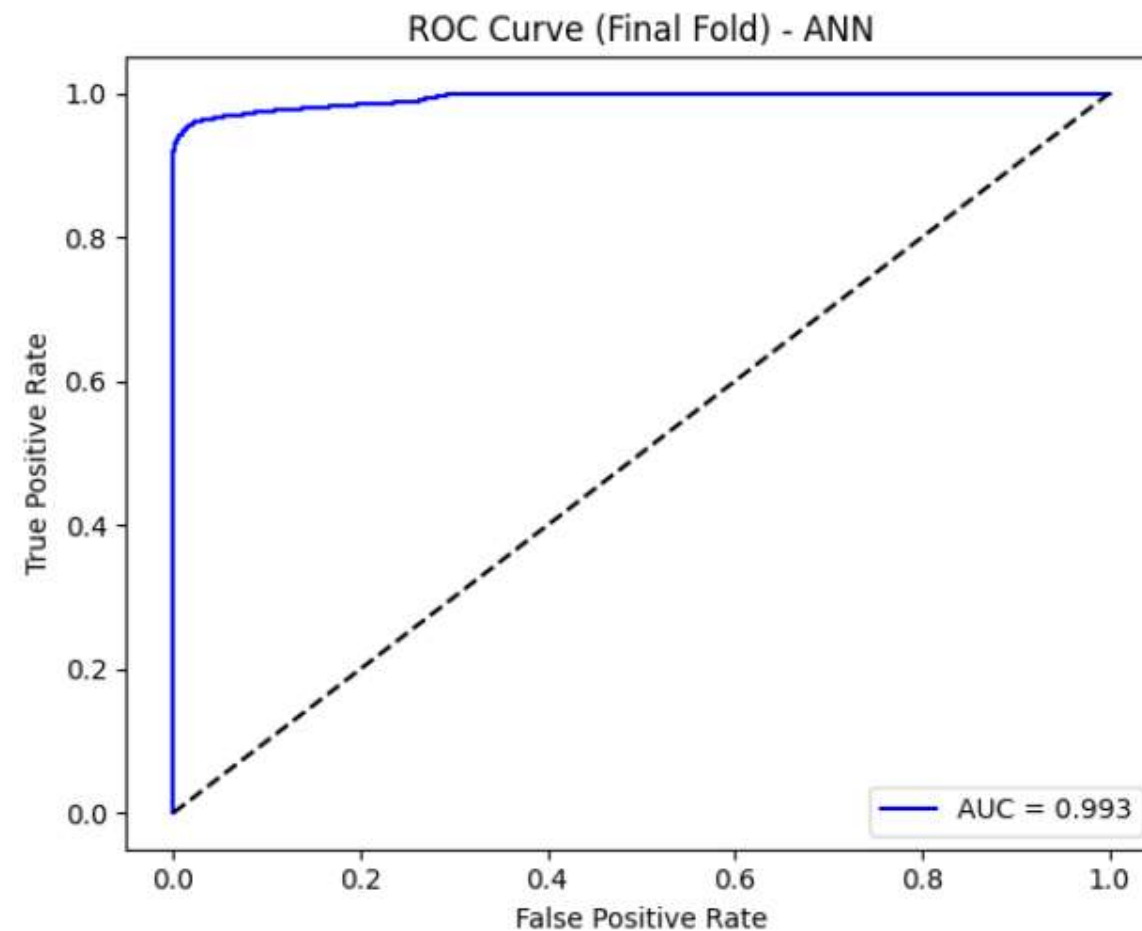
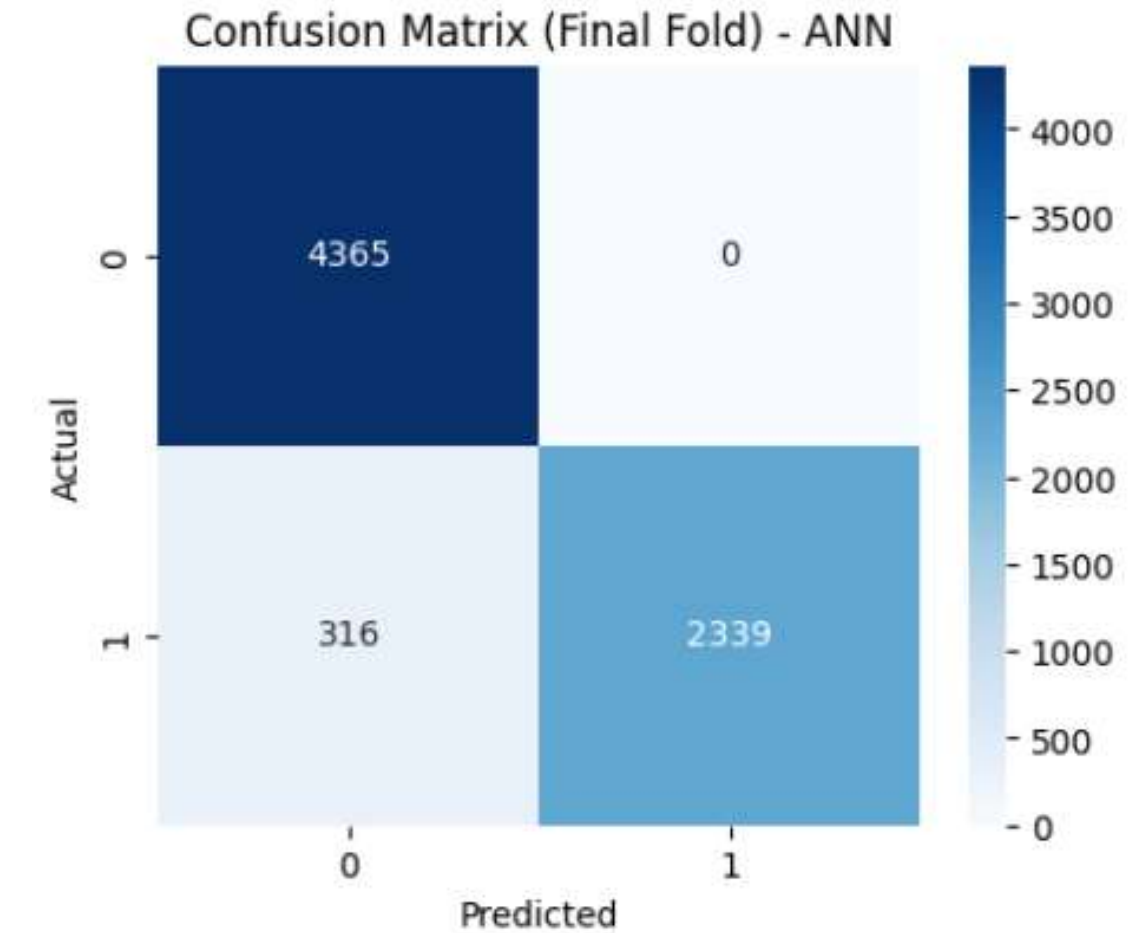
- Accuracy: **0.9673**
- F1-score: **0.9160**
- ROC-AUC: **0.9708**
- Precision: **0.9127**
- Recall: **0.9199**
- Precision@Top10%: **0.9536**
- CatBoost achieved **very strong classification performance** with balanced precision and recall values.
- The **ROC-AUC score (0.9708)** indicates excellent model discrimination capability between classes.
- Confusion matrix shows **minimal false positives and false negatives**, highlighting model reliability.
- **Time-fold AUC stability** remains consistently high, showing robustness across temporal splits.
- Overall, CatBoost performs **comparably to Random Forest and LightGBM**, maintaining high predictive accuracy and generalization.



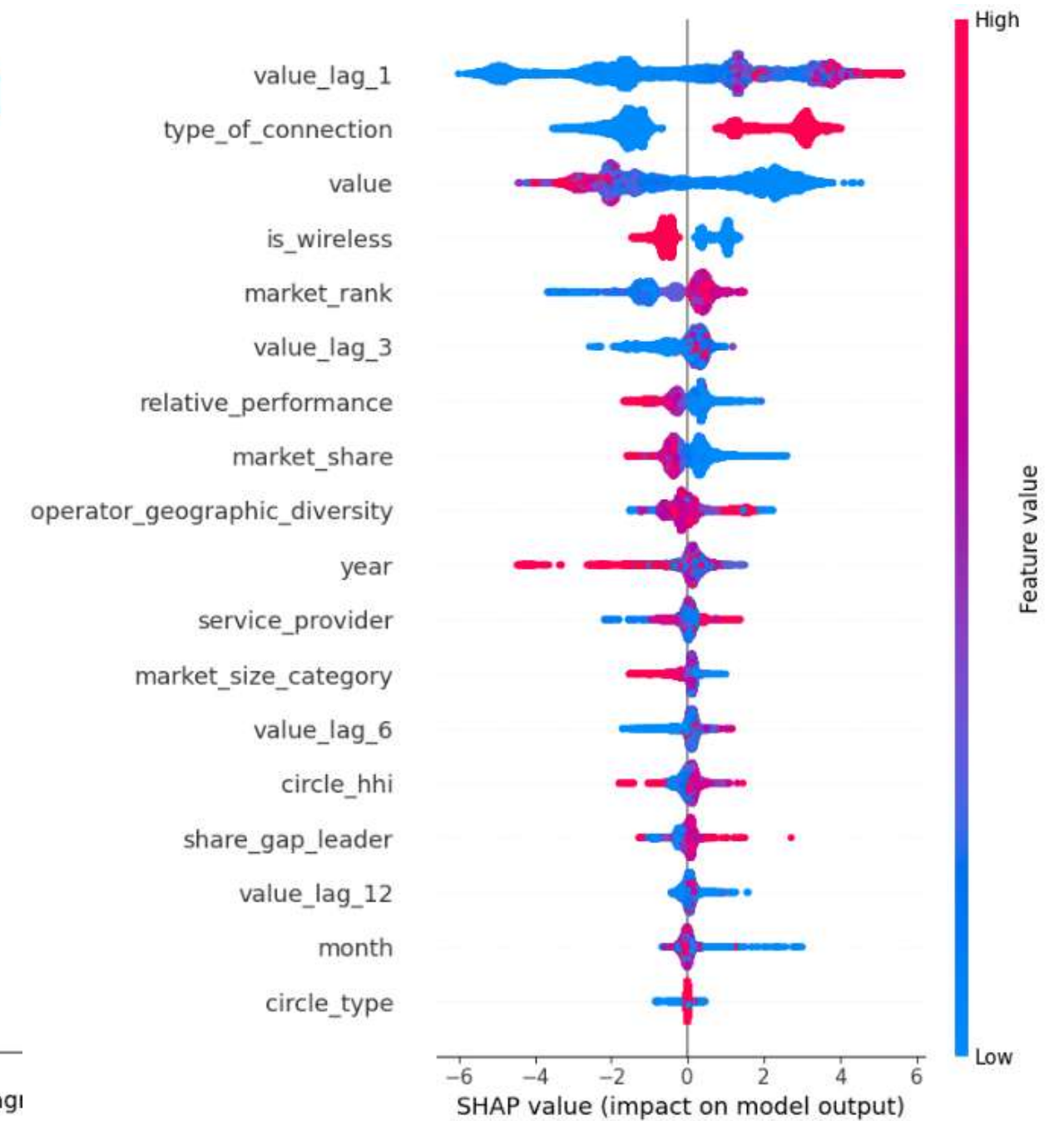
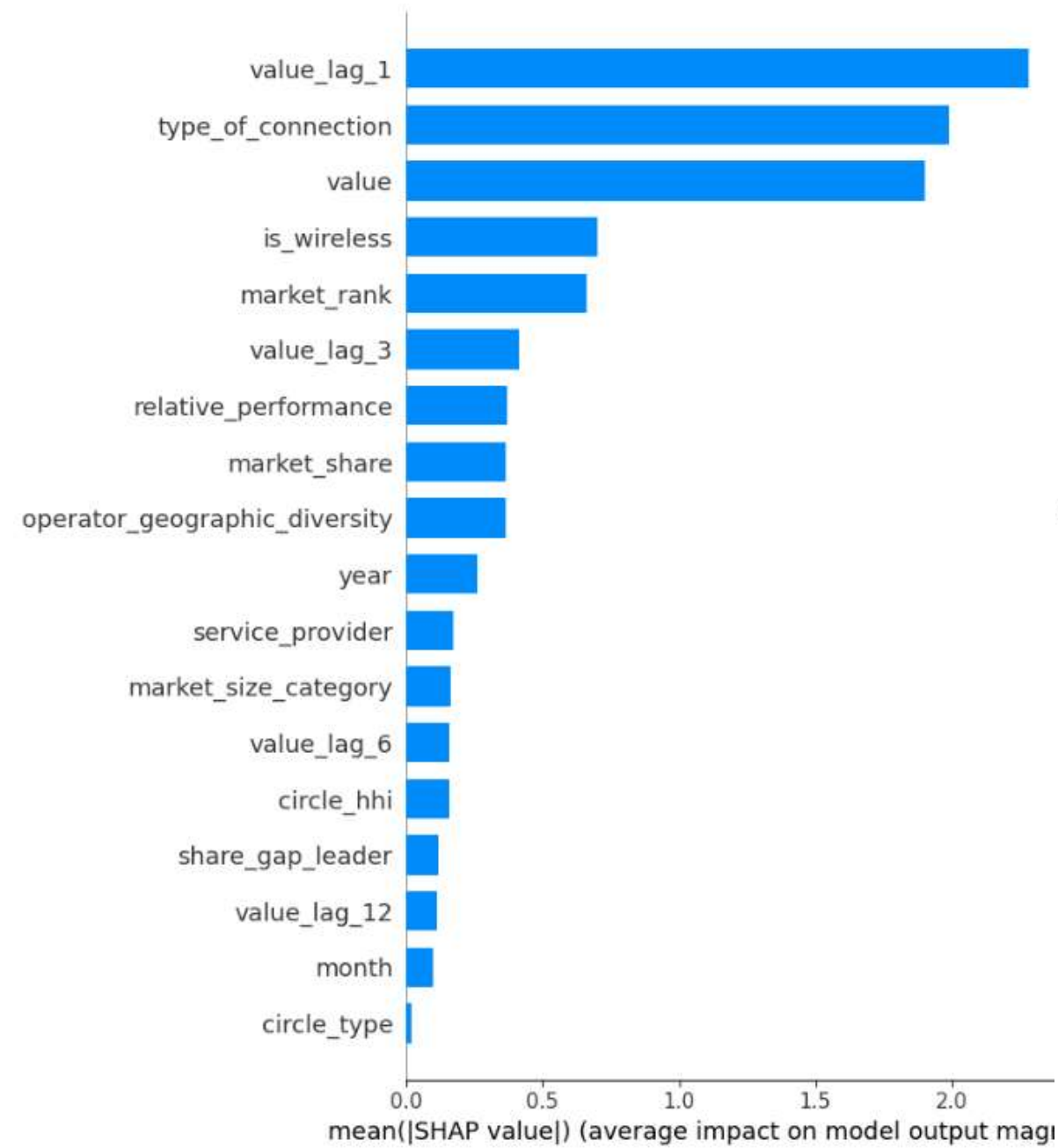
Artificial Neural Network (ANN) :

Metrics :

- Accuracy: **0.8988**
- F1-score: **0.8080**
- ROC-AUC: **0.9171**
- Precision: **0.8124**
- Recall: **0.8236**
- Precision@Top10%: **0.8100**
- The ANN model achieved a **good balance between precision and recall**, indicating stable classification performance.
- The **ROC-AUC of 0.9171** shows strong capability in distinguishing between classes.
- Slightly lower recall suggests some **false negatives**, but overall predictions remain consistent.
- **Confusion matrix** shows minimal false positives, meaning the model avoids overpredicting the positive class.
- Model performance is slightly lower than CatBoost, but still **reliable for real-world prediction scenarios**.



SHAP Analysis :



•**Top Drivers:** value_lag_1, type_of_connection and value are the most influential features shaping the model's predictions.

•**Temporal Impact:** Lag features (value_lag_1, value_lag_3, value_lag_6) show strong time-dependence — recent past values greatly affect outcomes.

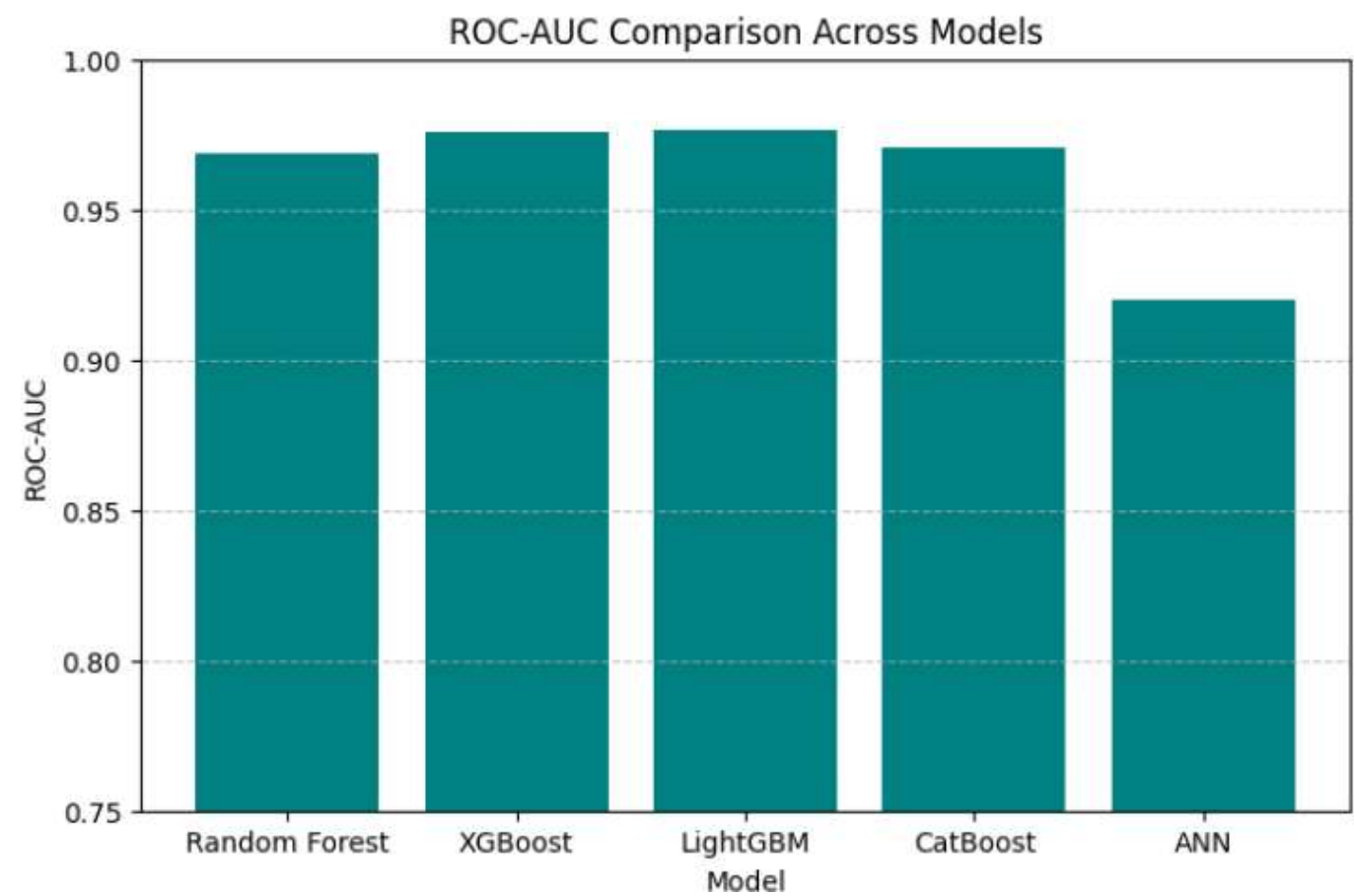
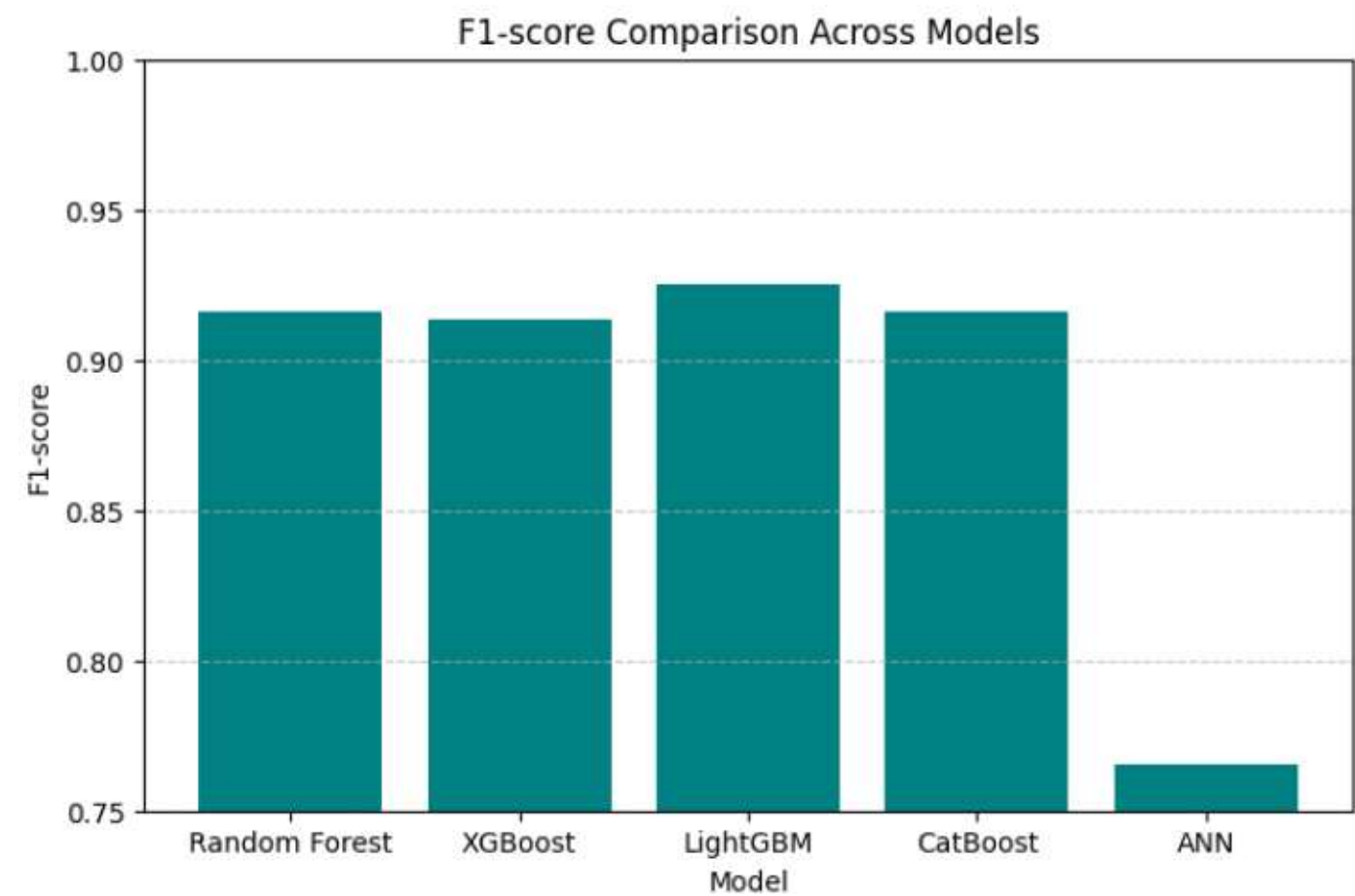
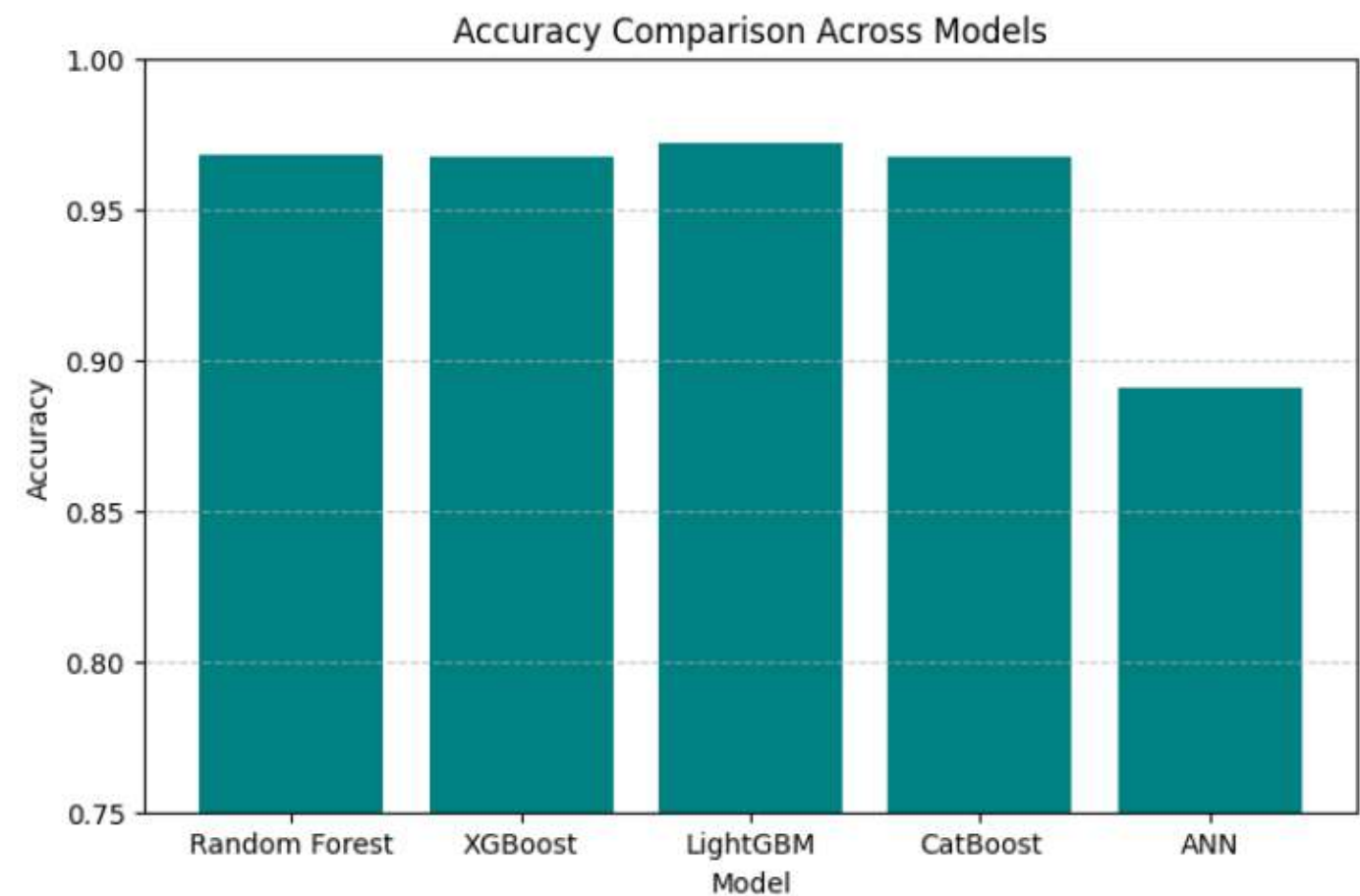
•**Connection & Market Factors:** type_of_connection, is_wireless, and market_rank significantly influence prediction behavior.

•**Moderate Influencers:** relative_performance, market_share, and operator_geographic_diversity add supporting context.

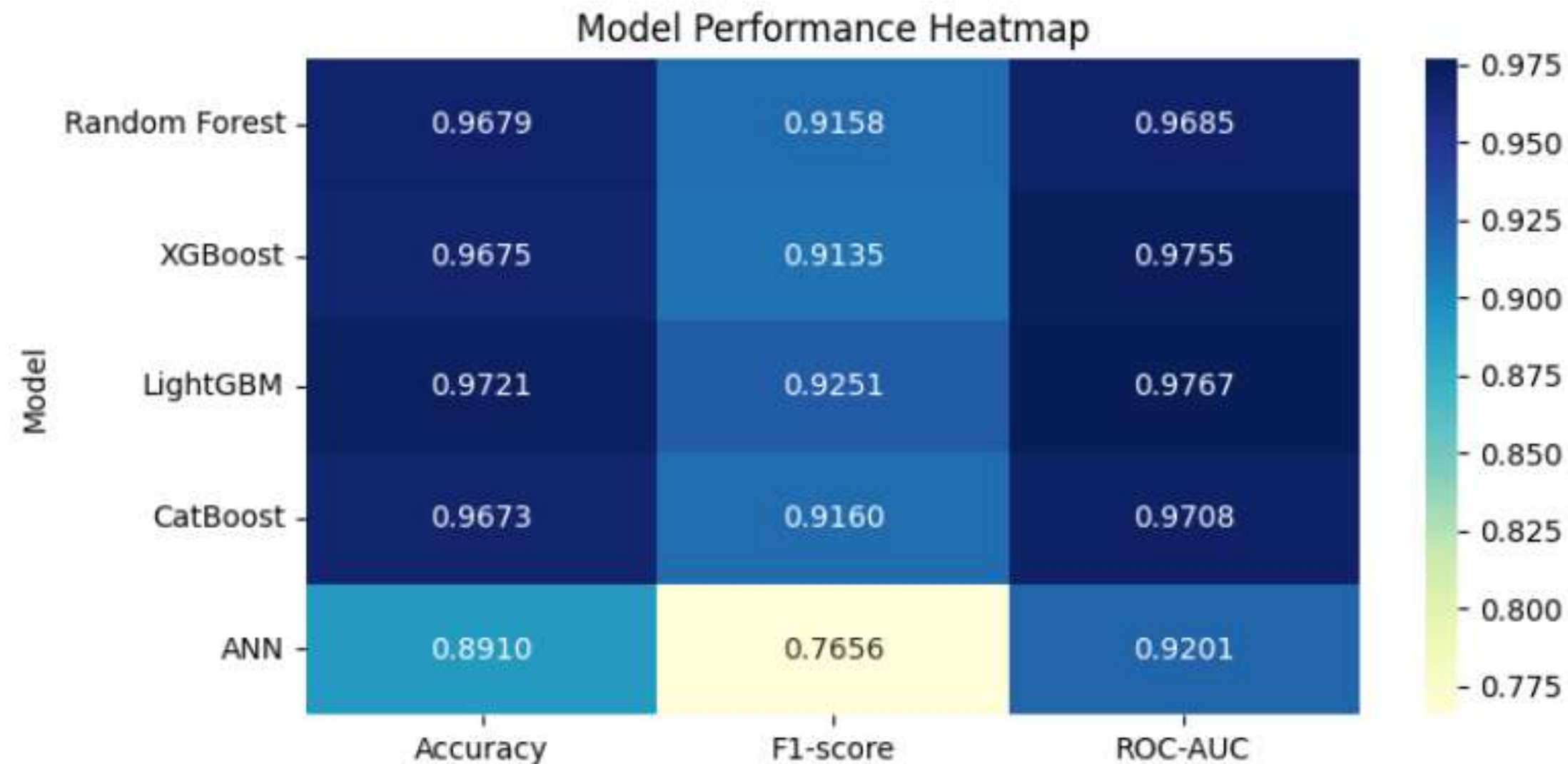
•**Low-Impact Features:** Variables like month, year, and circle_type have minimal effect on model output.

Overall Model Comparison

Model Comparison Charts :

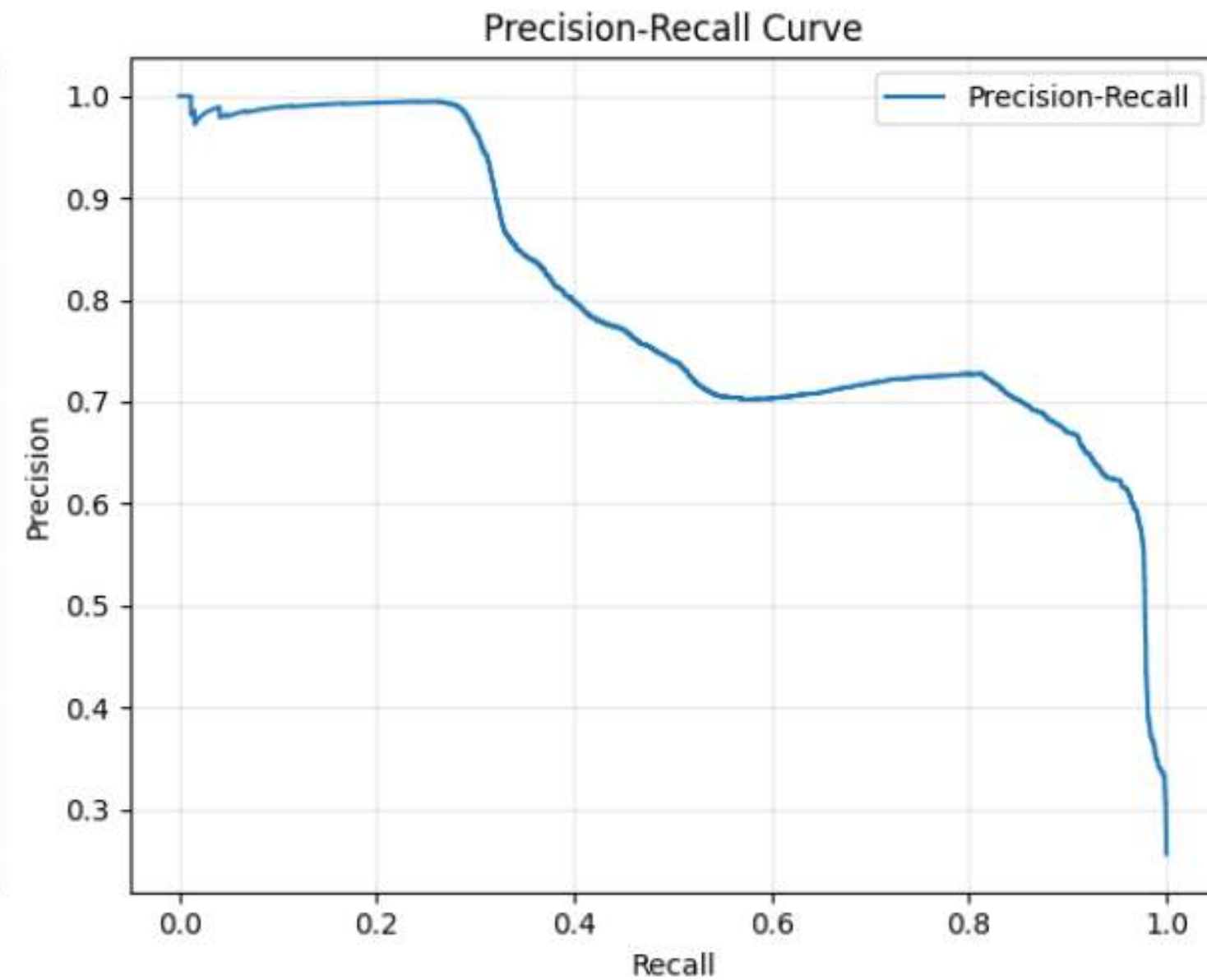
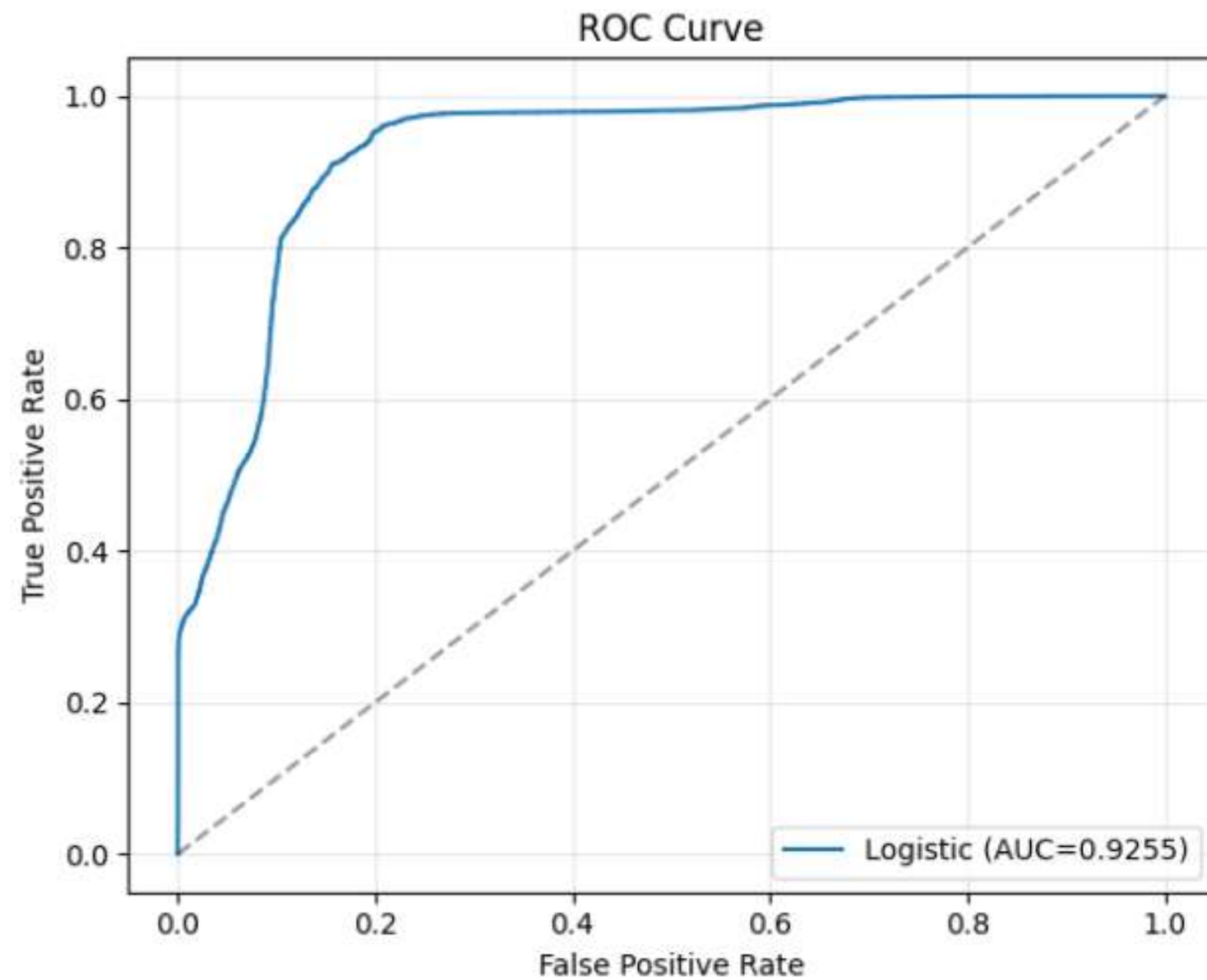


Model Performance Heatmap



- The darker the color, the better the model performance.
- **Light GBM** stands out as the overall best performer across all metrics (Accuracy, F1, ROC-AUC).
- **Random Forest, XG Boost, and Cat Boost** perform similarly and form a close second tier.
- **ANN** clearly underperforms, visible from its lighter color blocks — confirming weaker overall results.

ROC and Precision-Recall Curves



- The **ROC Curve** shows a strong performance with $AUC \approx 0.93$ — the curve hugs the top-left corner, meaning high true positive rate and low false positive rate.
- The **Precision-Recall Curve** maintains high precision for most recall values, indicating the model performs well even on imbalanced data.
- Together, these curves validate the model's reliability and robustness beyond simple accuracy metrics.

Churn Records

🔥 Top 100 High-Risk Churn Records

⬇️ 🔍 🗖️ Deploy

	service_provider	circle	value	predicted_churn_prob	churn_severity
9476	BSNL	Uttar Pradesh (East)	140938	1	High
9480	BSNL	Uttar Pradesh (East)	136164	1	High
9384	BSNL	Uttar Pradesh (East)	358167	1	High
9408	BSNL	Uttar Pradesh (East)	302460	1	High
9404	BSNL	Uttar Pradesh (East)	310533	1	High
4382	BSNL	Haryana	187134	1	High
9478	BSNL	Uttar Pradesh (East)	141044	1	High
8640	BSNL	Rajasthan	385204	1	High
9326	BSNL	Uttar Pradesh (East)	387814	1	High
7872	BSNL	Odisha	195006	1	High
...	...	...	...	...	...



Download Top 100 High-Risk Records as CSV

Conclusion & Business Impact

Key Takeaways:

- The developed **market churn analytics pipeline** achieved **>97% accuracy and ~0.98 ROC-AUC**, proving strong predictive capability.
- **Lag-based temporal features** and **competitive indicators (HHI, market rank, churn severity)** were the most influential drivers of churn.
- The model effectively highlights **high-risk operator–circle combinations**, enabling **targeted retention and marketing strategies**.

Business Value:

- Supports **data-driven decisions** for customer retention, market expansion, and pricing optimization.
- Provides a **dashboard-ready framework** for continuous monitoring of churn trends and competitive movements.
- Offers **ROI-focused insights** by identifying circles with the highest potential revenue loss or gain.

Thank You!!

